EUROSYSTEM LIQUIDITY, FINANCIAL DURATION AND RELATIVE ASSET VALUATION

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EUROSYSTEM LIQUIDITY, FINANCIAL DURATION AND RELATIVE ASSET VALUATION

TRABAJO DE FIN DE GRADO

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List of abbreviations

ADF	Augmented Dickey–Fuller
AIC	Akaike Information Criterion
APP	Asset Purchase Programme
BIC	Bayesian Information Criterion
CISS	Composite Indicator of Systemic Stress
DFR	deposit facility rate
ECB	European Central Bank
EMU	Economic and Monetary Union
€STR	euro short-term rate
HAC	heteroskedasticity and autocorrelation consistent
KPSS	Kwiatkowski–Phillips–Schmidt–Shin
MSCI	Morgan Stanley Capital International
OIS	overnight index swap
PEPP	Pandemic Emergency Purchase Programme
TPI	Transmission Protection Instrument
VAR	vector autoregression
VSTOXX	EURO STOXX 50 Volatility Index

Resumen

Este Trabajo de Fin de Grado analiza si las expansiones de liquidez del BCE y del Eurosistema alteran la valoración relativa de activos con distinta duración financiera. El marco teórico parte de la ecuación de valor presente y sostiene que, ante una compresión de los tipos de descuento o de las primas de riesgo, los activos cuyos flujos de caja esperados se concentran en horizontes más lejanos deberían experimentar una mayor revalorización. Empíricamente, esta hipótesis se aproxima mediante el diferencial de rentabilidad entre MSCI Europe Growth y MSCI Europe Value, utilizado como proxy imperfecta de duración financiera. El análisis emplea datos mensuales de enero de 2000 a febrero de 2026, una función de reacción del Eurosistema para modelizar la endogeneidad de la liquidez y proyecciones locales con errores estándar HAC de Newey–West para estimar respuestas dinámicas. Los resultados muestran que la liquidez responde parcialmente a condiciones financieras previas, lo que confirma la relevancia del problema de endogeneidad. Las innovaciones residuales no generan una mejora uniforme de las condiciones financieras agregadas: el CISS aumenta en determinados horizontes y la respuesta acumulada del EURO STOXX 50 es negativa. La evidencia de dependencia de régimen no es robusta. En cambio, el diferencial Growth menos Value responde positivamente y alcanza significatividad puntual en horizontes cortos y medios. La especificación de referencia, basada en la innovación residual de liquidez, alcanza su máximo en $h = 5$ ($\beta = 0.0092$, $p = 0.006$), mientras que la especificación alternativa basada en el crecimiento del balance lo alcanza en $h = 4$ ($\beta = 0.0101$, $p = 0.0022$), con estabilidad cualitativa en las principales especificaciones de robustez. En conjunto, los resultados son compatibles con un canal de duración financiera, aunque no constituyen identificación estructural y están limitados por el carácter paneuropeo de la proxy Growth/Value.

Palabras clave: Liquidez del Eurosistema, duración financiera, Growth versus Value, proyecciones locales, política monetaria no convencional

Abstract

This Bachelor's Thesis analyzes whether the ECB's and the Eurosystem's liquidity expansions alter the relative valuation of assets with different financial durations. The theoretical framework is based on the present value equation and argues that, in the event of a compression in discount rates or risk premiums, assets whose expected cash flows are concentrated in the more distant future should experience a greater increase in value. Empirically, this hypothesis is tested using the return differential between the MSCI Europe Growth and MSCI Europe Value indices, which serves as an imperfect proxy for financial duration. The analysis uses monthly data from January 2000 to February 2026, a Eurosystem reaction function to model the endogeneity of liquidity, and local projections with Newey–West HAC standard errors to estimate dynamic responses. The results show that liquidity responds partially to prior financial conditions, confirming the relevance of the endogeneity problem. Residual innovations do not lead to a uniform improvement in aggregate financial conditions: the CISS increases over certain horizons, and the cumulative response of the EURO STOXX 50 is negative. The evidence of regime dependence is not robust. In contrast, the Growth minus Value spread responds positively and reaches pointwise significance over short and medium horizons. The baseline specification, based on the residual liquidity innovation, peaks at $h = 5$ ($\beta = 0.0092$, $p = 0.006$), while the alternative specification based on balance sheet growth peaks at $h = 4$ ($\beta = 0.0101$, $p = 0.0022$), with qualitative stability across the main robustness specifications. Overall, the results are consistent with a financial duration channel, although they do not constitute structural identification and are limited by the pan-European nature of the Growth/Value proxy.

Keywords: Eurosystem liquidity, financial duration, Growth versus Value, local projections, unconventional monetary policy

1. Introduction

Over the last few decades, monetary policy in advanced economies has undergone a profound transformation. Central Bank actions are no longer limited to setting short-term interest rates, but now include unconventional instruments for providing liquidity, balance sheet expansion, asset purchases, and financial market stabilization. In Europe, this transformation has been particularly significant. The global financial crisis, the sovereign debt crisis, the period of negative interest rates and the COVID-19 pandemic cemented the European Central Bank and the Eurosystem as the central players in the eurozone´s liquidity architecture.

This undergraduate thesis analyzes this framework from the perspective of macrofinance and asset pricing. The main question is not simply whether liquidity expansions help stabilize financial markets during times of stress; but whether such liquidity can also alter the relative price structure within the asset market. In particular, this thesis examines whether the ECB´s and the Eurosystem´s liquidity expansions disproportionately favor those assets whose valuation depends to a greater extent on expected cash flows over longer time horizons

This insight is based on the standard asset pricing equation. Let P_t be the price of a financial asset in period t, CF_{t+j} the expected cashflow in period $t+j$, r_t^f the risk free rate and π_t the risk premium demanded by investors. The price can be expressed as the expected present value of its future cash flows:

$$P_t = \mathbb{E}_t \left[\sum_{j=1}^{\infty} \frac{CF_{t+j}}{(1 + r_t^f + \pi_t)^j} \right]$$

Defining the relevant discount rate as $R_t = r_t^f + \pi_t$, a liquidity expansion can affect financial valuations via two conceptually distinct channels. On the one hand, it can reduce the risk-free component if central bank action flattens the yield curve. On the other hand, it can reduce the risk premium if it improves financing conditions, alleviates financial stress, or prompts a shift in portfolio allocations toward higher-risk assets. In both cases, a reduction of R_t increases the present value of future cash flows, despite not all assets reacting with the same intensity.

An asset's sensitivity to the discount rate depends on the time distribution of its expected cash flows. Assets whose value depends more heavily on future cash flows have a longer financial duration and are therefore more sensitive to changes in the discount rate. This property allows us to link expansionary monetary policy to a relative valuation hypothesis: If liquidity compresses the relevant discount rate, assets with longer durations should benefit relatively more than those with shorter durations. Empirically, this study bridges that gap by comparing *Growth* portfolios, treated as assets with greater financial duration; and *Value* portfolios, treated as assets with lower financial duration.

1.1 Statement of the problem

The mian focus of this study arises from the interaction between three factors, the Eurosystem's liquidity supply, the impact of that liquidity on financial conditions, and the comparative worth of assets with varying durations. In times of stress, the ECB can expand its balance sheet, increase the liquidity available to the financial sector or launch asset purchase programs with the aim of preserving monetary transmission, reducing financial fragmentation and stabilizing markets. These interventions affect not

just the overall level of asset prices but can also change the relative prices in the market.

The analytical difficulty lies in the fact that central bank liquidity is not exogenous. The ECB and the Eurosystem do not expand their balance sheets at random; rather, they tend to do so precisely when financial conditions deteriorate, volatility increases, risk premiums widen or asset prices fall; thus, a contemporary correlation between liquidity expansion and stock market performance can be misleading. In the short term, liquidity may appear to be linked to periods of stress, not because it causes financial distress, but because it responds to it.

This endogeneity requires a distinction between the central bank's systematic response and the subsequent transmission of liquidity to the markets. A purely contemporary interpretation could blur the distinction between the moment of intervention and its resulting effect. Consequently, the examination necessitates a dynamic model that permits the investigation of how financial variables react over various timeframes and, as much as feasible, distinguishes the predictable aspect of the Eurosystem's reaction function from remaining liquidity disturbances.

The second aspect of the problem is the heterogeneity of financial assets. If expansionary monetary policy works by lowering discount rates and risk premiums, its effects should not be distributed evenly across all assets. Assets with cash flows expected further in the future should be more sensitive to a reduction in the discount rate than those whose value relies more heavily on current earnings, existing assets, or short-term cash flows. This difference in sensitivity introduces an intertemporal dimension into financial valuation allowing us to examine whether liquidity alters the relative price composition between *Growth* and *Value* assets.

Thus, the research question can be interpreted as a matter of macrofinancial transmission: how much does the liquidity offered by the ECB and the Eurosystem serve merely as a tool for overall stabilization; and, to what degree might it result in a relative repricing that favors assets with longer financial durations? This differentiation is important as it enables an evaluation of monetary policy not only regarding its impact on overall variables, but also concerning its consequences for the intertemporal framework of asset pricing.

1.2 Academic motivation and relevance

The academic motivation for this work lies at the intersection of monetary policy, financial conditions and asset pricing.The research on unconventional monetary policy has thoroughly explored the impacts of asset acquisitions, quantitative easing, the portfolio rebalancing mechanism, the signaling mechanism, and the reduction of risk premiums. Nonetheless, a substantial part of this body of work has concentrated on the overall effects on interest rates, spreads, volatility, or general stock market indices. This paper seeks to redirect attention, at least in part, to a more precise question: whether liquidity can influence the relative valuation of assets that have varying financial durations.

The eurozone is a particularly important case in this regard. Unlike a monetary union that is fully integrated in fiscal and financial terms, the euro area combines a shared monetary policy with national debt markets, variations in sovereign risk, episodes of financial fragmentation and a unique institutional structure. In this context, Eurosystem liquidity not only plays a role in monetary stabilization but helps to preserve financial transmission across different jurisdictions and market segments.

The period following the global financial crisis provides a valuable context for examining these dynamics. The combination of reduced interest rates, the expanding size of the Eurosystem's balance sheet, asset purchase initiatives, and persistent financial difficulties allows us to closely analyze how

liquidity, volatility, systemic stress, stock market dynamics, and the comparative values of different investment strategies interact. This analysis helps us understand if more liquidity pushes asset prices up and how it affects valuation incentives for assets with different cash flow timing.

From a macrofinancial perspective, expansionary monetary policy is relevant due to its dual effects. In the short to medium term, it can stabilize financial conditions, reduce risk premiums, and prevent chaotic deleveraging. However, if the ongoing compression of the discount rate systematically benefits assets with longer financial durations, it may also prompt a relative shift in valuation toward projects or companies whose value depends on more distant future expectations. While this scenario does not necessarily indicate the existence of a bubble or capital misallocation, it does provide a framework for analyzing a potential channel through which financial fragility may accumulate over time.

The significance of this analysis lies in offering a nuanced interpretation of monetary liquidity. Liquidity is considered not only as an aggregate stabilizing factor but also as a mechanism that influences the temporal structure of asset prices. This perspective connects the European Central Bank's actions to a central issue in modern macrofinance: the impact of monetary conditions on how markets price time, risk, and future cash flows.

1.3 Research question

The central question of the research is as follows:

> Do the ECB's and the Eurosystem's liquidity expansion measures favor the relative appreciation of assets with longer durations, approximated by *Growth* portfolios; over assets with shorter durations, approximated by *Value* portfolios?

This central question is divided into four sub-questions. First, we examine whether Eurosystem liquidity reacts endogenously to periods of financial stress, high volatility, or market deterioration. Second, we examine whether residual liquidity innovations are related to dynamic changes in financial conditions, as measured by variables such as systemic stress, implied volatility, market returns, or the term structure of interest rates. Third, we analyze whether liquidity transmission is conditioned by the financial regime, that is, whether the effects are more pronounced during episodes of stress than during periods of normality. Fourth, we compare whether the response of the *Growth minus Value* spread is consistent with the financial duration hypothesis.

The key empirical variable for this last dimension is defined as:

$$GV_t = r_t^{Growth} - r_t^{Value}$$

where r_t^{Growth} represents the *Growth* portfolio's return and r_t^{Value} *Value* portfolio returns. Under this convention, $GV_t > 0$ indicates that *Growth* outperforms *Value* in period t. If liquidity favors assets with longer durations, the expected response of this variable to an expansionary liquidity shock should be positive over the relevant time horizons.

1.4 Research hypothesis

The initial hypothesis of the study suggests that Eurosystem liquidity adjusts endogenously to financial conditions. Formally, if L_t denotes a measure of liquidity, $\ell_t = \log L_t$ its logarithmic transformation, and S_t a vector of financial stress variables, it is anticipated that the dynamics of liquidity can be

represented by a response function:

$$\Delta \ell_t = \alpha + \sum_{i=1}^{p} \rho_i \Delta \ell_{t-i} + \sum_{i=0}^{q} \gamma_i' S_{t-i} + u_t$$

This hypothesis implies that the expansion of liquidity should not be interpreted as a purely exogenous shock, but rather as a combination of a systematic response to financial distress and residual innovation not explained by the variables included in the reaction function.

The second hypothesis asserts that innovations in residual liquidity are associated with dynamic enhancements in financial conditions. When the Eurosystem's systematic actions are accounted for, liquidity innovations are expected to correspond with a later decrease in financial stress, a drop in implied volatility, a narrowing of risk premiums, or a rise in overall stock market returns. This hypothesis is explored via dynamic reactions by horizon, circumventing the constraint of deducing correlations in real-time.

The third hypothesis posits that the flow of liquidity is shaped by the financial system. During periods of significant stress, the supply of liquidity can have more pronounced effects, as it impacts a financial system characterized by balance sheet constraints, heightened risk aversion, strains on intermediation, and high risk premiums. Conversely, in normal times, the same expansion of liquidity may have a marginally reduced impact. This assumption requires that the calculated response vary according to the initial financial situation.

The fourth hypothesis, which is the core distinction of the study, posits that liquidity expansions relatively benefit assets with longer financial durations. If *Growth* assets derive a larger portion of their value from expected cash flows in the distant future, and *Value* assets depend to a greater extent on near-term cash flows or valuations based on current earnings, then a reduction in the discount rate should lead to a relative appreciation of *Growth*. Empirically, for a dynamic specification of the form:

$$GV_{t+h} = \alpha_h + \beta_h \varepsilon_t^{\ell} + \Gamma_h' X_{t-1} + u_{t+h}$$

where ε_t^{ℓ} denotes the residual liquidity innovation, the central hypothesis is:

$$\beta_h > 0$$

for a relevant set of horizons h. This condition would suggest that an expansion of liquidity is associated with higher relative returns for *Growth* relative to *Value*, consistent with the financial duration channel.

1.5 Contribution of the study

The study makes a threefold contribution. First, it applies a macrofinance question, one that is typically asked at the aggregate level, to the context of the eurozone: how does monetary liquidity affect asset valuations? Rather than focusing solely on the impact on a broad stock market index, the analysis incorporates the relative dimension among assets with different financial durations, as approximated by a comparison between *Growth* and *Value*.

Second, the study explicitly incorporates the issue of monetary policy endogeneity. Eurosystem liquidity is not treated as an exogenous variable, but rather as a measure that responds to adverse financial conditions. Consequently, the empirical design starts with a liquidity reaction function and

uses residual innovations to study the subsequent dynamic transmission. This methodological choice helps avoid an overly simplistic interpretation of the relationship between balance sheet expansion and asset prices.

Third, this paper links the literature on unconventional monetary policy with the theory of financial duration in equities. The comparison between *Growth* and *Value* is not presented as a perfect measure of duration, but rather as an empirical approximation of differential discount sensitivity. This approximation allows us to study a dimension that is difficult to observe directly: the way in which liquidity can affect the implied time structure of financial valuation.

This study adopts a cautious approach. It does not attempt to demonstrate that the European Central Bank (ECB) directly causes a bubble in *Growth* assets, nor does it seek to provide structural evidence of capital misallocation within the eurozone. Instead, the objective is to assess whether the observed dynamic patterns align with the hypothesis of transmission through the compression of discount rates and risk premiums, while accounting for the endogeneity of the central bank's reaction function and potential regime dependence.

A significant limitation of the analysis is that, in the absence of operational and sufficiently consistent time series for the MSCI EMU Growth and MSCI EMU Value indices, the *Growth versus Value* section uses the MSCI Europe Growth and MSCI Europe Value indices as an empirical approximation. This choice allows us to capture the cross-sectional dimension of discount sensitivity, but introduces a geographical constraint: MSCI Europe does not strictly represent the eurozone, but rather developed Europe. Consequently, the results of this section should be interpreted as European evidence consistent with the financial duration channel, not as a pure measure of relative valuation within the euro area.

1.6 Structure of the paper

The paper is organized into nine chapters. Chapter 2 describes the euro-area liquidity architecture, and Chapter 3 develops the theoretical framework linking discount-rate compression to financial duration. Chapter 4 reviews the relevant literature and identifies the research gap. Chapters 5 and 6 present the data, variable construction, and econometric identification strategy. Chapter 7 reports the empirical results, Chapter 8 discusses their macrofinancial interpretation and limitations, and Chapter 9 concludes by answering the research question and outlining future research.

2. The Eurozone's Liquidity Architecture

2.1 The ECB and the Eurosystem as providers of liquidity

The euro area's liquidity architecture can be defined as the set of institutions, operational instruments, collateral rules, refinancing programs, and asset portfolios through which the European Central Bank and the national central banks of the Eurosystem supply, absorb, and redistribute central bank liquidity within the euro area's financial system. This architecture is not merely a technical mechanism for monetary policy implementation. In a monetary union comprising multiple national banking systems, differentiated sovereign markets, and heterogeneous financial structures, the provision of liquidity fulfills a central macrofinancial function; enabling the transmission of monetary policy, stabilizing the money market, preserving banking intermediation and reducing the risk of financial fragmentation.

The European Central Bank (ECB) establishes the overall direction of monetary policy, which the Eurosystem, comprising the ECB and the national central banks of euro area countries, implements in a decentralized manner. The primary objective of this framework is to ensure that decisions made by the Governing Council influence short-term interest rates, bank lending conditions, and capital markets, thereby affecting spending, investment, inflation, and broader economic activity. A distinction must be made between central bank liquidity, referring to reserve balances and liquid assets managed by the central bank, and market liquidity, defined as the ability to trade financial assets without causing significant price fluctuations. These two forms of liquidity are interconnected: an increase in central bank liquidity can alter asset demand dynamics, affect liquidity and risk premiums, and shape the term structure of interest rates.

From an operational standpoint, the ECB implements its monetary policy through three main categories of instruments: open market operations, standing facilities, and minimum reserve requirements (European Central Bank, 2024b). Open market operations form the core of day-to-day liquidity management. Among these, the main refinancing operations, typically with a one-week maturity; and the longer-term refinancing operations, typically with a three-month maturity, stand out (European Central Bank, 2024c). Both allow the Eurosystem to provide liquidity to the banking system against eligible assets as collateral. Standing facilities, on the other hand, allow banks to obtain overnight liquidity through the marginal lending facility or to deposit excess liquidity through the deposit facility. Reserve requirements oblige credit institutions to maintain a certain amount of funds in their accounts with national central banks, thereby helping to stabilize the structural demand for reserves.

Formally, if R_t represents the reserves held by credit institutions in the Eurosystem and MR_t represents the required reserve ratio, a simplified measure of excess liquidity can be expressed as:

$$EL_t = R_t - MR_t$$

This concept reflects the fundamental notion that excess liquidity arises when the banking system maintains central bank balances exceeding minimum reserve requirements. Economically, $EL_t > 0$ signifies that the banking system possesses surplus liquidity, which may remain on deposit with the Eurosystem, circulate within the interbank market, or indirectly affect portfolio allocation and broader financial conditions. The European Central Bank (ECB) defines excess liquidity as the funds in the banking system that persist after banks have fulfilled their minimum reserve requirements (European Central Bank, 2023).

The significance of this framework for the present study lies in the fact that the Eurosystem's liquidity is not merely an accounting variable. The expansion of the central bank's balance sheet simultaneously alters both the assets and liabilities of the Eurosystem. On the asset side, there is an increase in credit operations to the banking system or in holdings of securities acquired through purchase programs. On the liability side, bank reserves, deposits from financial institutions, and other monetary liabilities increase. This accounting relationship allows us to link monetary policy to the discount channel and to asset pricing. When the Eurosystem purchases assets or provides abundant financing, it alters the supply of safe assets available to the private sector, changes the relative price of risk, and can compress the required yields on financial assets of different maturities.

2.2 From Conventional Monetary Policy to Unconventional Monetary Policy

The ECB's conventional monetary policy was based primarily on controlling official interest rates and managing bank liquidity through refinancing operations. Under normal operating conditions, the monetary authority could guide short-term interest rates through the corridor formed by the marginal lending facility and the deposit facility, with the main refinancing operations serving as the central mechanism for providing liquidity. The monetary policy signal was initially transmitted to the money market and, from there, to the rest of the yield curve, bank lending, financing conditions for businesses and households, and aggregate demand.

The 2008 global financial crisis fundamentally disrupted the established monetary policy framework. Surrounded by interbank mistrust, elevated risk premiums, declining collateral quality and significant funding pressures, the primary concern shifted from setting the official interest rate to ensuring the financial system's capacity to absorb and redistribute liquidity. In response, the Eurosystem implemented fixed-rate full allotment procedures, broadened access to funding, reinforced the collateral framework, and lengthened the maturity of liquidity operations. As a result, conventional monetary policy, which traditionally emphasized the price of liquidity, was increasingly supplemented by measures targeting the quantity, maturity, and distribution of liquidity.

The following European sovereign debt crisis further accelerated this transformation. Unlike the United States or China, the eurozone operates with a single currency but multiple sovereign issuers, distinct national banking systems and partially segmented financial markets. As a result, a single monetary policy decision may be transmitted unevenly across member states, especially if sovereign spreads widen, the interbank market fragments, or banks in certain jurisdictions face increased funding constraints. Therefore, unconventional monetary policy in the euro area should be interpreted not only as a response to the effective lower bound on interest rates but also as an effort to address challenges related to policy transmission, market fragmentation, and financial segmentation.

In this context, unconventional measures included longer-term refinancing operations, credit-linked refinancing operations, forward guidance, negative rates on the deposit facility, and asset purchase programs. The common feature of these tools is that they shift the emphasis from the official short-term interest rate to the broader set of financial conditions relevant to households, firms, banks, and governments. Monetary policy now operates through several simultaneous channels: the expectations channel, the term structure channel, the collateral channel, the banking channel, the portfolio rebalancing channel, and the risk premium channel.

The literature on asset purchases has shown that the effects of quantitative easing are not limited

to the assets purchased directly by the central bank. Krishnamurthy and Vissing-Jorgensen (2011) argue that purchase programs can operate through various channels, including signaling, duration, a shortage of safe assets, liquidity, and a reduction in risk premiums. This idea is particularly relevant to the present study because it allows us to link the Eurosystem's liquidity expansion to a change in relative prices across assets. If asset purchases reduce risk-free rates or risk premiums, assets with longer durations should experience a greater percentage change in their valuation. This intuition will be formally developed in the theoretical framework below.

2.3 Balance sheet expansion, excess liquidity, and asset purchase programs

The expansion of the Eurosystem's balance sheet is the most visible accounting manifestation of unconventional monetary policy. When the central bank provides long-term funding to the banking system or purchases securities in the market, the size of its assets increases. At the same time, liabilities to the financial system increase, particularly bank reserves and deposits in the deposit facility; from the banks' perspective, these reserves are liquid and safe assets; from the Eurosystem's perspective, they are liabilities created as a counterpart to its monetary operations.

This relationship can be represented in simplified form using the balance equation:

$$\Delta A_t^{CB} = \Delta R_t + \Delta O_t$$

where ΔA_t^{CB} denotes the change in central bank assets, ΔR_t the change in bank reserves, and ΔO_t the change in other Eurosystem liabilities. While this identity does not capture the full complexity of the balance sheet, it clarifies the fundamental mechanism by which asset purchases and refinancing operations expand liquidity in the financial system by creating central bank liabilities.

Excess liquidity arises when the amount of reserves supplied by the central bank exceeds the banking system's minimum needs. In an environment of abundant reserves, banks do not compete intensely for liquidity in the interbank market, and very short-term rates tend to hover near the deposit facility rate. In such an environment, the deposit facility plays a central role as an anchor for monetary rates. The operational framework announced by the ECB in 2024 confirms precisely that, in a context of gradual balance sheet normalization and a progressive decline in excess liquidity, the Eurosystem will continue to aim for short-term rates to evolve close to the deposit facility, preserving the clarity of the monetary policy signal (European Central Bank, 2024a).

Asset purchase programs constitute a separate channel for balance sheet expansion, distinct from credit operations with the banking system. Rather than supplying liquidity against collateral with a fixed maturity, the Eurosystem acquires financial securities, primarily government and corporate bonds, and retains them in its monetary policy portfolio. These purchases decrease the availability of such assets in the private sector, increase their prices, and reduce their yields. Additionally, they prompt investors to rebalance portfolios toward assets with longer durations, higher risk, or lower liquidity. This mechanism is essential for understanding how liquidity influences both the aggregate level and the relative composition of financial prices.

The Asset Purchase Programme (APP) was the ECB's main purchase program prior to the pandemic. It included various subprograms, such as purchases of covered bonds, asset-backed securities, public sector bonds, and corporate bonds. The ECB decided to end net purchases under the

APP in July 2022 and discontinued reinvestments as of July 2023 (European Central Bank, 2026a). This development is significant because it marks the transition from a phase of active balance sheet expansion to a phase of gradual reduction through maturities. In empirical terms, this implies that Eurosystem liquidity should not be treated as a monotonically increasing variable, but rather as a series with phases of expansion, stabilization, and relative contraction.

The Pandemic Emergency Purchase Programme (PEPP) was introduced as an extraordinary response to the COVID-19 crisis. Unlike previous programs, its defining feature was the flexibility of purchases over time, across asset classes, and across jurisdictions, with the aim of preserving a uniform transmission of monetary policy under conditions of severe stress (European Central Bank, 2026c). The ECB concluded net purchases under the PEPP at the end of March 2022 and discontinued reinvestments at the end of 2024 (European Central Bank, 2026c). The flexibility of the PEPP is crucial for analyzing the euro area's liquidity architecture, as it shows that asset purchases served not only as a tool for aggregate stimulus but also as a mechanism to contain financial fragmentation.

Balance sheet expansion and excess liquidity are relevant for three primary reasons. First, they provide an observable measure of the intensity of the Eurosystem's monetary intervention. Second, they influence the term structure of interest rates and risk premiums, which are critical components of the discount rate applied to future cash flows. Third, they can trigger portfolio rebalancing, increasing demand for assets more sensitive to the discount rate, such as growth assets. Consequently, Eurosystem liquidity is conceptualized as a macrofinancial variable that simultaneously affects both aggregate financial conditions and relative asset prices.

2.4 APP, PEPP, and Financial Stabilization

The APP and the PEPP constitute two distinct phases of the ECB's asset purchase policy. The APP was introduced in response to persistently low inflation, interest rates near the effective lower bound, and the necessity of providing additional stimulus to the euro area economy. Its primary objectives were to reduce medium- and long-term yields, ease financing conditions, and promote the convergence of inflation toward the price stability target. In contrast, the PEPP was implemented in the context of the pandemic shock, market dislocation, a sharp increase in sovereign spreads, and heightened risks to monetary transmission. While both programs utilized central bank asset purchases as their core mechanism, their macrofinancial roles differed significantly.

Empirical evidence regarding the APP indicates that the ECB's asset purchases significantly influenced financial prices. Altavilla et al. (2015) estimate that both the announcement and anticipation of the APP reduced yields across multiple market segments, with more pronounced effects on long-term maturities and higher-risk assets. This result is central to the current analysis. When asset purchases compress long-term yields and risk premiums, monetary transmission operates through the factors that determine the intertemporal valuation of assets. The reduction in the discount rate does not affect all assets uniformly; assets with longer financial durations exhibit a greater price response to a given change in the discount rate.

The PEPP, for its part, was more explicitly designed to stabilize the economy. The pandemic led to a sharp rise in uncertainty, a deterioration in market liquidity, and the risk of divergence in national financial conditions. In this context, the flexibility of the PEPP allowed the Eurosystem to adjust purchases in response to observed tensions. This flexibility should not be interpreted as a minor operational detail, but rather as an essential feature of the European liquidity architecture. In

an incomplete monetary union, liquidity provision must respond not only to the aggregate cycle, but also to the spatial distribution of financial stress.

Financial stabilization through asset purchases operates through several channels. The first channel is the interest rate channel, through which purchases lower the yields on the acquired assets and, via arbitrage, affect the rest of the yield curve; the second channel is the portfolio channel, whereby investors who sell assets to the central bank replace those assets with other financial instruments, raising their prices and lowering their expected returns, the third channel is the signaling channel, insofar as the purchases signal a more accommodative future monetary policy stance; and the fourth channel is the liquidity and market functioning channel, which is particularly relevant during periods of stress, when the central bank's presence can reduce liquidity premiums and restore market depth.

These channels are consistent with the central hypothesis of this study. If asset purchases and the expansion of liquidity reduce r_t^f, the risk-free component of the discount rate, or π_t, the risk premium demanded by investors, then the aggregate discount rate can be expressed as:

$$R_t = r_t^f + \pi_t$$

and asset valuation is affected by:

$$P_t = \mathbb{E}_t \left[\sum_{j=1}^{\infty} \frac{CF_{t+j}}{(1 + R_t)^j} \right]$$

The subsequent theoretical section formally derives this expression. At this stage, it is important to note that the Eurosystem's liquidity architecture establishes the institutional conditions under which R_t can be compressed. The financial stabilization channel is not independent of the valuation channel. The same intervention that reduces financial stress may also increase the present value of future cash flows and modify the relative returns on assets with varying durations.

However, this interpretation calls for caution. The APP and the PEPP should not be viewed as purely exogenous shocks. The ECB does not expand its balance sheet at random, but rather acts in response to low inflation, market failures, deteriorating financial conditions, or the risk of fragmentation; therefore, a contemporary correlation between liquidity expansion and a decline in financial assets may reflect the central bank's reaction to stress, not a negative effect of liquidity. This consideration justifies the empirical strategy of this study of first estimating a Eurosystem reaction function and then analyzing the dynamic responses to residual liquidity innovations.

2.5 Liquidity, Financial Fragmentation, and Monetary Transmission in the Eurozone

Financial fragmentation is one of the key distinguishing features of the eurozone compared to other currency areas. In a fully integrated monetary union, a monetary policy decision should be transmitted relatively uniformly to all agents subject to the same currency. However, the eurozone combines a single monetary policy with national banking systems, sovereign markets, and credit conditions that may diverge during periods of stress. When sovereign spreads widen in a disorderly manner, when the interbank market becomes segmented by jurisdiction, or when banks in some countries face higher funding costs, the transmission of monetary policy becomes uneven.

The overnight money market plays a central role in this transmission mechanism. The ECB empha-

sizes that the overnight money market is the starting point of the monetary transmission mechanism, because expectations regarding official interest rates determine expected overnight rates, and these serve as an anchor for the entire yield curve (Eisenschmidt et al., 2018). If this first link breaks down, the rest of the transmission mechanism may also deteriorate. In that case, the same official rate can lead to very different financial conditions across countries, banks, or economic sectors.

Fragmentation also affects the interpretation of aggregate liquidity. An increase in total liquidity within the Eurosystem does not necessarily guarantee a uniform distribution of liquidity across jurisdictions or institutions. There may be an aggregate abundance of reserves while, at the same time, localized funding strains or elevated risk premiums persist in certain markets. Therefore, the analysis of liquidity in the euro area must incorporate variables such as systemic stress, implied volatility, sovereign spreads, or fragmentation indicators. In the empirical design of this study, this consideration justifies the use of variables such as CISS, VSTOXX, EURO STOXX 50 returns, yield curve slope, and financial regime variables.

The ECB's institutional response to this problem has been to develop instruments specifically designed to preserve monetary transmission. The PEPP introduced flexibility in cross-jurisdictional purchases, precisely to prevent a common shock from disrupting transmission. Subsequently, the Transmission Protection Instrument (TPI) was approved in 2022 to counteract disorderly and unjustified market dynamics that threaten the transmission of monetary policy in the euro area (European Central Bank, 2022). The existence of this instrument reveals an important conceptual distinction: not every central bank intervention should be interpreted as a change in the aggregate monetary stance. Some interventions aim to preserve the transmission channel, preventing fragmentation from hindering the stance decided by the Governing Council from reaching the real economy in a uniform manner.

This distinction is relevant to this study because Eurosystem liquidity can serve a dual function. On the one hand, it can act as a monetary policy tool, easing or tightening aggregate financial conditions. On the other hand, it can act as a mechanism to stabilize and safeguard the transmission mechanism during periods of stress. This dual function introduces endogeneity: liquidity increases precisely when financial conditions deteriorate. Therefore, the relationship between liquidity and asset prices can be ambiguous. In periods of stress, an expansion of liquidity may coincide with stock market declines, increased volatility, or widening spreads, not because liquidity causes financial deterioration, but because it responds to it.

The euro area's liquidity architecture should therefore be understood as a system for providing liquidity, stabilizing markets, and safeguarding monetary transmission. Its institutional specificity lies in the fact that it operates within a monetary union comprising multiple sovereign and banking markets. This characteristic means that the liquidity provided by the ECB and the Eurosystem can have potentially complex effects on asset prices. It can affect not only the aggregate level of valuation through discount rates and risk premiums, but also the relative composition of that valuation across assets with different financial durations. This is the point of connection between this institutional chapter and the subsequent chapters of the study: the liquidity architecture creates the operational framework within which we empirically examine whether liquidity expansions favor Growth assets relative to Value assets.

3. Theoretical framework: liquidity, discount, and financial duration

3.1 Asset Valuation and Discount Rate

The theoretical starting point of this study is the central concept of asset valuation: the price of a financial asset can be interpreted as the expected present value of the future cash flows that the asset will generate. In its most general form, this relationship can be expressed using the stochastic discount factor approach, according to which the current price depends on future payments weighted by a factor that incorporates the time value of money, macro risk, and risk aversion (Cochrane, 2005). If P_t represents the price of the asset in period t, CF_{t+j} the expected cash flow at $t+j$, and $M_{t,t+j}$ the stochastic discount factor between t and $t+j$, then:

$$P_t = \mathbb{E}_t \left[\sum_{j=1}^{\infty} M_{t,t+j} CF_{t+j} \right]$$

To link this insight to the financial time horizon, it is helpful to work with a simplified model in which expected cash flows are discounted using an aggregate discount rate R_t:

$$P_t = \mathbb{E}_t \left[\sum_{j=1}^{\infty} \frac{CF_{t+j}}{(1+R_t)^j} \right].$$

In equity valuation, R_t should not be interpreted solely as a risk-free rate. The relevant discount rate incorporates the risk-free rate, the risk premium demanded by investors, the liquidity premium, uncertainty regarding future earnings and compensation for exposure to adverse economic cycles. In simple terms:

$$R_t = r_t^f + \pi_t$$

where r_t^f represents the risk-free component and π_t captures the aggregate risk premium. This decomposition allows us to directly link monetary policy to asset pricing. A liquidity expansion can raise P_t if it reduces r_t^f, if it compresses π_t, or if it acts simultaneously on both components.

The above equation shows that the price of an asset can vary for two main reasons. First, due to revisions in expected cash flows, that is, changes in $\mathbb{E}_t(CF_{t+j})$; and second, due to changes in the discount rate applied to those cash flows, that is, changes in R_t. The central hypothesis of this paper focuses primarily on this second dimension: Eurosystem liquidity can alter the intertemporal discounting of future cash flows, affecting assets with different financial durations to varying degrees. The log-linear present-value framework separates unexpected equity returns into revisions to expected future cash flows and revisions to expected future returns, or discount rates (Campbell & Shiller, 1988). Campbell and Vuolteenaho extend this decomposition to the cross-section of equity returns, distinguishing between *cash-flow news* and *discount-rate news* and showing that these two components carry different prices of risk (Campbell & Vuolteenaho, 2004).

3.2 Financial duration and discount sensitivity

Financial duration measures the sensitivity of an asset's value to changes in the discount rate. In fixed income, duration can be interpreted as the weighted average maturity of cash flows. In equities, the concept is less directly observable because cash flows are uncertain, non-contractual, and potentially indefinite. However, the economic logic is analogous: an asset has a longer financial duration when a larger proportion of its value depends on expected cash flows in the distant future (Dechow et al., 2004).

Using the valuation equation, it can be shown that the percentage sensitivity of the price to the discount rate depends on the financial life of the asset. In simplified form, the first-order approximation is:

$$\Delta \ln P_t \approx -D_t \Delta R_t$$

where D_t represents the modified financial duration of the asset. This expression captures the essential intuition of the chapter. If $\Delta R_t < 0$, the price increases. Furthermore, the larger D_t is, the greater the percentage change in price in response to a given change in the discount rate. The formal derivation of this relationship, including the definition of the weights $w_{t,j}$, the implied Macaulay duration, and the modified duration, is presented in Appendix A.4.

This property links monetary liquidity and relative valuation. If the ECB and the Eurosystem inject liquidity and this injection drives down the relevant discount rate, assets with longer durations should react more strongly than those with shorter durations. Therefore, unconventional monetary policy can affect not only the aggregate level of financial prices but also the relative composition of valuations within the market.

3.3 Liquidity Transmission Channels to the discount and portfolios

Eurosystem liquidity can affect the discount rate through several channels. The first is the interest rate channel. When the central bank cuts official interest rates, signals a more accommodative future policy path, or purchases long-term assets, it can reduce the risk-free component of the discount rate:

$$\Delta r_t^f < 0$$

The second channel is the compression of risk premiums. During periods of financial stress, the provision of liquidity can improve market functioning, reduce balance sheet constraints, curb forced sales, lower liquidity premiums, and reduce the compensation required for risk. In that case:

$$\Delta \pi_t < 0$$

Since $R_t = r_t^f + \pi_t$, the total change in the discount rate can be written as:

$$\Delta R_t = \Delta r_t^f + \Delta \pi_t$$

If the expansion of liquidity reduces the risk-free rate, the risk premium, or both, then $\Delta R_t < 0$. Using the duration approach, this reduction increases the present value of future cash flows, with a greater impact on assets with longer financial durations.

The third channel is portfolio rebalancing. When the central bank purchases safe or long-duration

assets, it reduces the amount of duration risk that the private sector must absorb. In the presence of preferred habitats, balance sheet constraints, or imperfect substitutability among assets, this net reduction in supply can compress returns and shift demand toward other financial assets (Krishnamurthy & Vissing-Jorgensen, 2011; Vayanos & Vila, 2021). Evidence on the ECB's purchase programs shows that the Asset Purchase Program affected financial returns and prices in the eurozone, providing an empirical basis for linking asset purchases, risk premiums, and financial valuation (Altavilla et al., 2015).

These three channels are not independent. In practice, an expansion of the Eurosystem's balance sheet can simultaneously influence expected interest rates, term premiums, market liquidity, risk appetite, and portfolio composition. Accordingly, this study does not attempt to structurally isolate each channel, but instead assesses whether the dynamic response of assets aligns with a compression of the aggregate discount. This distinction is significant: the financial duration channel does not require separate identification of each component of R_t, but rather demonstrates that assets most sensitive to the discount rate respond more strongly to liquidity expansions.

3.4 Growth vs. Value as an Empirical Approach to Financial Duration

The distinction between *Growth* and *Value* is used as an empirical approximation of the difference in financial duration between assets. *Growth* portfolios typically consist of companies whose valuations depend more heavily on expectations of future growth and earnings over the long term. *Value* portfolios, on the other hand, are typically more closely associated with current earnings, dividends, existing assets, book value, or lower valuation multiples. From an intertemporal valuation perspective, this difference implies that *Growth* should be more sensitive than *Value* to changes in R_t.

Formally, for a portfolio $i \in \{G, V\}$, the approximate sensitivity to the discount rate can be written as:

$$\Delta \ln P_t^i \approx -D_t^i \Delta R_t^i$$

If *Growth* has a greater financial life than *Value*, then:

$$D_t^G > D_t^V$$

Given a common or highly correlated decrease in the discount rate, $\Delta R_t < 0$, the expected difference in relative valuation is:

$$\Delta \ln P_t^G - \Delta \ln P_t^V \approx -\left(D_t^G - D_t^V\right) \Delta R_t > 0$$

This inequality underpins the cross-sectional hypothesis of the study: if liquidity compresses the relevant discount rate, *Growth* assets should outperform *Value* assets. Therefore, the main empirical variable in the cross-sectional section is defined as:

$$GV_t = r_t^{Growth} - r_t^{Value}$$

Under this convention, $GV_t > 0$ indicates that *Growth* outperforms *Value* in period t. In local projections, the central hypothesis is expressed as a positive response of the cumulative differential *Growth minus Value* to an expansionary liquidity shock:

$$GV_{t,t+h} = \alpha_h + \beta_h \varepsilon_t^{\ell} + \Gamma_h' X_{t-1} + u_{t+h}, \qquad \beta_h > 0$$

This interpretation should be made with caution. *Growth* and *Value* do not differ solely in terms of financial duration. They may also differ in sector composition, geographic exposure, currency, cyclical sensitivity, earnings expectations, dividend yield, capital intensity, and exposure to technological shocks. Furthermore, the study uses MSCI Europe Growth and MSCI Europe Value as the available European proxy, not MSCI EMU Growth and MSCI EMU Value. Therefore, the differential GV_t should be interpreted as an imperfect but defensible proxy for relative financial duration in European developed markets, not as a pure measure of the eurozone nor as an exact structural measure of duration.

In summary, the theoretical framework of this study is articulated as a conditional causal chain. First, a liquidity expansion can compress r_t^f, π_t, or both. Second, this compression reduces R_t. Third, the decline in R_t increases the present value of future cash flows. Fourth, this increase is more pronounced for assets with longer financial durations. Fifth, if *Growth* exhibits a longer duration than *Value*, then a liquidity expansion should be associated with a relative improvement in *Growth* compared to *Value*. Formally:

$$\Delta L_t > 0 \quad \Rightarrow \quad \Delta R_t < 0 \quad \Rightarrow \quad \Delta \ln P_t^G - \Delta \ln P_t^V > 0,$$

$$\text{given that } D_t^G > D_t^V.$$

The preceding derivation isolates the discount-rate channel by holding expected cash flows fixed. This conditioning is essential. A decline in observed interest rates does not mechanically imply a relative revaluation of *Growth*, because the same rate movement may reflect adverse news about expected growth or risk rather than a pure decline in required returns. In the Campbell–Shiller return decomposition, unexpected returns reflect both *cash-flow news* and *discount-rate news*; consequently, the net relative response of *Growth* and *Value* depends on the joint revision of expected cash flows and discount rates (Campbell & Shiller, 1988; Campbell & Vuolteenaho, 2004). This distinction is particularly relevant because *Growth* firms can be interpreted as longer-horizon claims and tend to covary more strongly with discount-rate shocks, whereas *Value* firms tend to covary more strongly with cash-flow shocks (Campbell & Vuolteenaho, 2004; Lettau & Wachter, 2007). Recent evidence further indicates that the transmission from real interest rates to equity valuations is strongest for the pure-discounting component of rate movements, while changes driven by expected-growth or risk shocks generate much weaker, and potentially negative, valuation effects (Gormsen & Lazarus, 2026). The duration hypothesis should therefore be interpreted as conditional: it is most likely to operate when liquidity provision, risk-premium compression, or portfolio rebalancing lowers required returns without an offsetting deterioration in the expected cash flows of *Growth* relative to those of *Value*.

This proposition does not assert strong structural causality. Eurosystem liquidity is endogenous and typically increases during periods of financial stress. For this reason, the subsequent empirical analysis distinguishes between the central bank's systematic response and residual liquidity innovations. Nevertheless, the theoretical framework provides a clear prediction: if the financial duration channel is operative, expansionary liquidity innovations should, over certain horizons, be associated with superior relative returns for *Growth* compared to *Value*.

4. State of the Art

4.1 Unconventional monetary policy and asset prices

Research on unconventional monetary policy has emerged to explain how central banks influence financial conditions when the official short-term interest rate approaches the effective lower bound. Traditionally, monetary transmission operates through the control of short-term rates, the shaping of expectations about future rates, bank credit, and aggregate demand. However, in response to the global financial crisis, the European sovereign debt crisis, and the pandemic, central banks adopted additional instruments, including asset purchases, long-term refinancing operations, forward guidance, negative interest rates, and programs to preserve monetary transmission.

The key contribution of this literature is to show that unconventional monetary policy can affect asset prices even when the official short-term interest rate remains constant. Asset purchases can alter the term structure of interest rates, reduce term premiums, ease liquidity constraints, induce portfolio rebalancing, and alter the risk premium. Consequently, the effect of unconventional monetary policy is not limited to the money market but extends to sovereign bonds, corporate credit, equities, exchange rates, and other financial assets (Gagnon et al., 2011; Krishnamurthy & Vissing-Jorgensen, 2011; Swanson, 2021).

In terms of valuation, these results are consistent with the view that an asset's price represents the expected present value of its future cash flows:

$$P_t = \mathbb{E}_t \left[\sum_{j=1}^{\infty} \frac{CF_{t+j}}{(1+R_t)^j} \right]$$

Unconventional monetary policy can affect P_t through two general channels. The first acts on the risk-free rate, r_t^f, through the expected path of official interest rates and the compression of safe-haven asset yields. The second acts on the risk premium, π_t, by reducing financial stress, improving market liquidity, and increasing risk appetite. If the aggregate discount rate is defined as:

$$R_t = r_t^f + \pi_t$$

Therefore, a reduction in either of these two components can increase the present value of future cash flows.

The literature on quantitative easing has emphasized that the effects of asset purchases depend on financial frictions. In a frictionless model with complete markets and perfect arbitrage, the composition of the central bank's balance sheet would have limited effects on relative prices. However, when preferred habitats, balance sheet constraints, investor segmentation, variable risk aversion, or imperfectly substitutable assets exist, the relative amount of assets held by the private sector can affect required returns. This intuition is formalized in the preferred habitat literature, according to which the relative supply of assets with different maturities can influence the term structure of interest rates if arbitrageurs do not completely eliminate risk-adjusted yield differentials (Vayanos & Vila, 2021).

This point is relevant to the present study as it enables a shift from an aggregate interpretation of liquidity to a relative one. If unconventional monetary policy compresses the discount rate, its effect may not be uniform across assets. Assets whose valuation relies more heavily on distant cash flows are expected to exhibit greater percentage sensitivity to changes in R_t. This insight links the

literature on unconventional monetary policy with research on financial duration and relative valuation, particularly the distinction between Growth and Value assets.

4.2 ECB Asset Purchases and Financial Transmission in the Eurozone

The literature on the ECB examines how the Eurosystem's asset purchase programs affected financial markets in the eurozone. Unlike economies with a single fiscal authority and a unified sovereign market, the eurozone combines a single monetary policy with multiple sovereign markets, national banking systems, and heterogeneous risk premiums. Therefore, the ECB's asset purchases should be interpreted not only as instruments of aggregate stimulus but also as mechanisms for preserving monetary transmission.

A key reference is Altavilla et al. (2015), which analyzes the effects of the Asset Purchase Program on financial prices in the eurozone. Their work is particularly useful because it links the ECB's purchase program to various transmission channels, including yield compression, risk premiums, and portfolio effects. This literature supports the argument that changes in Eurosystem liquidity can affect not only narrow monetary variables but also financial asset prices and yield spreads.

The European case introduces an additional dimension: financial fragmentation. During the sovereign debt crisis, the ECB's official interest rate was transmitted unevenly across countries due to divergences in sovereign spreads, banking risk, and financing conditions. In this context, asset purchases and liquidity operations helped reduce market tensions and preserve the unity of monetary policy. The objective was not only to ease aggregate financial conditions but also to prevent fragmentation from hindering the uniform transmission of the monetary stance.

The Pandemic Emergency Purchase Programme reinforced this logic. Its flexible design permitted adjustments to purchases over time, across jurisdictions, and among asset classes. This flexibility demonstrates that unconventional monetary policy in the euro area extends beyond simple balance sheet expansion. The composition, timing, and distribution of interventions are central elements of the liquidity architecture. Accordingly, Eurosystem liquidity should be analyzed as an endogenous variable, linked to episodes of financial stress and stabilization requirements.

The European literature provides a foundation for examining the relationship between Eurosystem liquidity and financial prices. However, most existing research focuses on sovereign yields, spreads, bank lending conditions, or aggregate asset prices. This study adopts a complementary perspective by investigating whether European liquidity is also associated with changes in the relative composition of stock market valuations, specifically between Growth and Value stocks.

4.3 Portfolio rebalancing, signaling, and transmission channels

The literature on the transmission channels of asset purchases distinguishes several conceptually different channels. The first is the signaling channel, according to which purchases convey information about the future direction of monetary policy. If investors interpret a purchase program as a signal of lower interest rates for longer, the yield curve may shift downward even before the purchases materially alter the supply of assets. Bauer and Rudebusch (2014) show that the signaling channel may be relevant in explaining the effects of bond purchases on interest rates.

The second channel is the duration channel, or scarcity channel. When the central bank purchases long-duration assets, it reduces the private sector's duration risk. If investors demand compensation

for holding duration, and if assets with different maturities are not perfect substitutes, the net reduction in supply can compress term premiums. This insight aligns with the literature on preferred habitat (Vayanos & Vila, 2021).

The third channel is the liquidity channel. During periods of stress, central bank intervention can improve market functioning, reduce liquidity premiums, and stabilize prices. This channel is particularly important when markets experience forced selling, balance sheet constraints, or a deterioration in collateral quality. In such circumstances, the central bank's balance sheet expansion can serve as a mechanism for absorbing risk and providing liquidity.

The fourth channel is portfolio rebalancing. Investors who sell assets to the central bank receive liquidity or deposits, but may seek other assets to restore their expected returns. This process shifts demand toward assets not purchased directly, such as corporate bonds, equities, higher-risk assets, or longer-duration assets. Krishnamurthy and Vissing-Jorgensen (2011) emphasizes that purchase programs can operate through multiple channels, including duration, liquidity, security, and default risk.

Formally, if B_t^{CB} represents the central bank's holdings of assets and B_t^{priv} represents the amount of risk assets or duration held by the private sector, we can write:

$$B_t^{priv} = B_t^{total} - B_t^{CB}$$

If the premium demanded by the private sector is positively related to the amount of risk it must assume, then:

$$\pi_t = \psi B_t^{priv} \qquad \psi > 0$$

Replacing:

$$\pi_t = \psi(B_t^{total} - B_t^{CB})$$

and differentiating with respect to B_t^{CB}:

$$\frac{\partial \pi_t}{\partial B_t^{CB}} = -\psi < 0$$

The intuition is straightforward: the greater central bank absorption of assets reduces the premium demanded by the private sector, assuming the presence of frictions or imperfect substitutability among assets. This relationship explains how liquidity can be transmitted to relative prices. If the decline in π_t reduces the aggregate discount rate, assets with longer financial durations are expected to benefit more significantly.

This literature allows us to situate the Growth-minus-Value spread within the monetary transmission mechanism. Here, Growth is not interpreted as a purely accounting or sectoral category, but rather as a proxy for assets whose valuation depends relatively more on distant cash flows. Value is interpreted as an approximation of assets with a greater weighting of present cash flows, existing assets, or current earnings. If portfolio rebalancing and the compression of premiums favor assets with longer durations, the differential:

$$GV_t = r_t^{Growth} - r_t^{Value}$$

is expected to respond positively to liquidity-expanding measures.

4.4 Financial Stress, Volatility, and Systemic Indicators

A significant portion of the macro-financial literature examines how to measure financial stress episodes and how these episodes influence the transmission of monetary policy. In the case of the eurozone, a key reference is Holló et al. (2012), who developed the Composite Indicator of Systemic Stress (CISS). This indicator synthesizes information from different segments of the financial system and captures episodes of systemic stress more broadly than a single market variable.

The CISS is useful for this study in two ways. First, it enables control for the initial financial state in the Eurosystem's reaction function. If liquidity increases in response to systemic stress, ignoring this relationship can result in misinterpretations. A contemporaneous correlation between liquidity expansion and declining asset prices may indicate that the central bank is responding to financial deterioration rather than liquidity causing it. Second, the CISS can be used to define high-stress regimes and to examine whether liquidity transmission depends on the financial system's initial state.

The literature on implied volatility serves a complementary function. Indicators such as the VSTOXX capture volatility expectations in European equities and can be interpreted as measures of uncertainty, risk aversion, or market stress. While the CISS provides a systemic, multi-sector measure, implied volatility more directly captures perceived stress in the equity market. Therefore, both variables can be used complementarily: the CISS as a broad systemic indicator and VSTOXX as a measure of equity-specific stress.

From an econometric perspective, the presence of financial stress introduces nonlinearity. The transmission of liquidity expansion may differ between normal and high-stress periods. Under normal conditions, a marginal expansion of liquidity may have limited effects if markets are already functioning properly. In contrast, during episodes of stress, the same expansion can reduce liquidity premiums, alleviate balance sheet constraints, and restore market depth. Therefore, this study proposes specifications with regime interaction:

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h \varepsilon_t^{\ell} + \theta_h (\varepsilon_t^{\ell} \times D_t^{Stress}) + \lambda_h D_t^{Stress} + \Gamma_h' X_{t-1} + u_{t+h}$$

In this expression, β_h measures the response under normal conditions, while $\beta_h + \theta_h$ measures the response under stress. This formulation is consistent with the literature that views unconventional monetary policy as particularly relevant when financial frictions are present. The central bank not only modifies relative prices through discount rates, but can also alter the intensity of the transmission depending on the initial financial regime.

4.5 Financial duration, equity duration, and relative valuation: Growth vs. Value

The equity duration literature provides the theoretical basis for interpreting Growth and Value as assets with different sensitivities to the discount rate. In fixed income, duration measures the weighted average maturity of cash flows and the price's sensitivity to changes in the required rate of return. In equities, applying the concept is more complex because cash flows are uncertain, non-contractual, and potentially infinite. However, Dechow et al. (2004) shows that an implied equity duration measure can be constructed as a natural extension of bond duration.

The intuition is that companies whose value depends more on expected profits in the distant future have greater financial duration. In contrast, companies with greater weightings of current profits, dividends, or existing assets have relatively shorter financial durations. This distinction is empirically approximated by comparing Growth and Value stocks. Growth companies typically have higher expectations for future growth and therefore place greater weight on future cash flows. Value companies tend to be more anchored in current fundamentals and to trade at lower valuation multiples.

Formally, for a portfolio *i*, the price can be expressed as:

$$P_t^i = \mathbb{E}_t \left[\sum_{j=1}^{\infty} \frac{CF_{t+j}^i}{(1 + R_t^i)^j} \right]$$

The approximate sensitivity of the price to the discount rate is:

$$\Delta \ln P_t^i \approx -D_t^i \Delta R_t^i$$

If Growth has a longer duration than Value:

$$D_t^G > D_t^V$$

Therefore, in the event of a common or highly correlated decline in the discount rate:

$$\Delta R_t^G \approx \Delta R_t^V < 0$$

the result is:

$$\Delta \ln P_t^G - \Delta \ln P_t^V \approx -\left(D_t^G - D_t^V \right) \Delta R_t > 0$$

However, interpreting Growth versus Value as an empirical approach to financial duration should be treated with caution. The relationship between interest rates, discount rates, and relative valuation is not mechanical because movements in risk-free rates may simultaneously reflect changes in growth expectations, risk aversion, term premiums, or aggregate financial conditions. In this sense, a decline in interest rates does not necessarily imply an immediate relative appreciation of Growth assets if that decline is associated with a deterioration in expectations of future earnings. Recent literature on equity duration and interest rates emphasizes precisely this distinction: the sensitivity of long-duration stocks is more pronounced in response to pure discount rate shocks than to gross interest rate movements, which may mix information on expected growth, risk, and market premiums (Gormsen & Lazarus, 2026).

This caution aligns with the classic literature, which breaks equity risk into cash flow shocks and discount rate shocks. Campbell and Vuolteenaho show that the exposure of equity portfolios cannot be reduced to a single aggregate beta, but rather depends on whether the shocks affect cash flow expectations or discount rates (Campbell & Vuolteenaho, 2004). Complementarily, Lettau and Wachter develop an explanation of the value premium based on the time structure of cash flows, distinguishing between firms whose value depends relatively more on near-term cash flows and firms with a greater weight of distant cash flows (Lettau & Wachter, 2007). Therefore, the use of Growth minus Value as a proxy for financial duration is theoretically defensible, but it should be understood as a noisy empirical approximation, not as a perfect structural measure of duration.

This relationship provides the formal basis for the central hypothesis of the Growth versus Value

section. Recent literature on equity term structures reinforces this intuition by examining how stock and portfolio prices can be decomposed according to the maturity of expected cash flows (Giglio et al., 2024). Although this literature does not imply that every difference between Growth and Value is due to duration, it does provide a rationale for interpreting part of the spread as a difference in discount rate sensitivity.

From an operational perspective, the MSCI Europe Growth and MSCI Europe Value indices provide an institutional and investable approach to the cross-cutting Growth versus Value dimension in European developed markets. MSCI's methodology classifies the Growth style using variables linked to expected and observed earnings and sales growth, while the Value style is defined by ratios tied to book value, expected short-term earnings, and dividend yield (MSCI, 2021, 2026a, 2026b). This construction is consistent with the concept of financial duration: Growth includes companies whose valuations depend more on expectations of future expansion, while Value is more anchored in current or near-term fundamentals. However, this equivalence should be understood as an empirical approximation rather than a pure structural measure of duration.

Nevertheless, using Growth versus Value as a proxy calls for caution. Growth and Value portfolios also differ in terms of sector composition, exposure to technological shocks, capital intensity, expected returns, dividend policies, currency, regulation, and the economic cycle. Therefore, the spread:

$$GV_t = r_t^G - r_t^V$$

is not a pure measure of financial duration. It is an empirical proxy. Its usefulness lies in the fact that it captures a cross-sectional dimension that is reasonably consistent with discounting theory: Growth should, on average, be more sensitive to changes in R_t than Value. This limitation does not invalidate the approach, but it requires interpreting the results as evidence consistent with the duration channel rather than as a perfect identification of structural duration.

4.6 Endogeneity of monetary policy and the central bank's reaction function

A particularly important line of research for this study is the examination of monetary policy endogeneity. Central banks do not adjust their balance sheets at random. They respond to inflation, economic activity, financial stress, credit conditions, volatility, and market functioning. Therefore, any empirical analysis linking liquidity and asset prices must acknowledge that liquidity may increase precisely when financial conditions deteriorate.

The literature on the central bank's reaction function shows that financial markets can influence monetary policy behavior. Cieslak and Vissing-Jorgensen (2021) examines the relationship between stock market declines and monetary policy easing in the United States. Although their evidence pertains to the Federal Reserve, its value for this paper is conceptual: it helps justify why monetary expansion may appear contemporaneously associated with poor financial conditions, as the central bank responds to them. In the case of the eurozone, this consideration is even more relevant given financial fragmentation and episodes of sovereign stress.

The problem can be formally stated. If L_t represents the liquidity of the Eurosystem, a naive specification might estimate:

$$y_{t+h} = \alpha_h + \beta_h \Delta \ell_t + u_{t+h}$$

However, if $\Delta \ell_t$ is a response to financial stress S_t, then:

$$\Delta \ell_t = \alpha + \sum_{i=1}^{p} \rho_i \Delta \ell_{t-i} + \sum_{i=0}^{q} \gamma_i' S_{t-i} + \varepsilon_t^{\ell}$$

In that case, $\Delta \ell_t$ contains a systematic reaction component and a residual component. Interpreting β_h as a direct causal effect of liquidity would be problematic, because part of the variation in liquidity reflects precisely the financial conditions that also affect y_{t+h}.

For this reason, this paper adopts a two-step strategy. First, it estimates a reaction function for Eurosystem liquidity. Second, it uses the residual innovation ε_t^{ℓ} in local projections. The Local Projections methodology, proposed by Jordà (2005), allows for the estimation of dynamic responses by horizon without imposing the entire dynamic structure of a VAR. For each horizon *h*, the general specification is:

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h \varepsilon_t^{\ell} + \Gamma_h' X_{t-1} + u_{t+h}$$

This methodology is well-suited for this study because it allows us to examine the dynamic trajectories of financial conditions, stock market returns, and the growth-minus-value spread following liquidity innovations. Furthermore, it facilitates the incorporation of regime interactions, lagged controls, and robust standard errors that account for autocorrelation and time overlap.

The literature on monetary identification also notes that residual innovations do not necessarily equate to pure exogenous shocks. Identifying monetary shocks typically requires narrative strategies, high-frequency data, structural constraints, or external instruments (Gürkaynak et al., 2005; Swanson, 2021). This paper takes a more cautious approach. It does not claim to identify structural monetary shocks, but rather conditional liquidity innovations. The empirical interpretation should be formulated as a dynamic association consistent with theoretical channels, not as definitive structural causality.

4.7 Research gap and contribution of the study

The existing literature provides the key building blocks of the argument. Asset pricing theory shows that prices depend on the expected present value of future cash flows. The literature on unconventional monetary policy shows that asset purchases and balance sheet expansion can suppress interest rates and risk premiums. The literature on the ECB shows that the APP and the PEPP had significant effects on financial markets and monetary transmission in the eurozone. The literature on systemic stress provides tools such as the CISS to capture financial regimes. The equity duration literature allows us to interpret Growth and Value as portfolios with different discount sensitivities. Finally, the literature on the central bank's reaction function warns that liquidity is endogenous and must be modeled with caution.

However, these bodies of literature are not typically integrated into a single framework applied to the eurozone's liquidity architecture and the relative valuation of Growth versus Value. Much of the research on the ECB's asset purchases focuses on sovereign bonds, interest rates, spreads, or aggregate financing conditions. Much of the literature on Growth versus Value is developed within the context of cross-sectional asset pricing, without explicitly linking the relative dynamics of invest-

ment styles to central bank liquidity. And much of the literature on unconventional monetary policy analyzes aggregate effects, without studying in detail how discount compression can alter the relative composition of stock market valuations.

The contribution of this thesis lies in connecting these dimensions. The study examines whether the ECB's and the Eurosystem's liquidity expansions, by lowering discount rates and risk premiums, are associated with a relative appreciation of assets with longer financial durations, approximated by Growth, relative to assets with shorter financial durations, approximated by Value. This contribution is relevant because it shifts the question from whether liquidity raises aggregate asset prices to a more nuanced question: whether liquidity alters the relative price structure within the financial market.

The central hypothesis can be summarized as follows:

$$\Delta L_t > 0 \quad \Rightarrow \quad \Delta R_t < 0 \quad \Rightarrow \quad \Delta \ln P_t^G - \Delta \ln P_t^V > 0$$

given that:

$$D_t^G > D_t^V$$

Empirically, this prediction is tested by:

$$GV_{t,t+h} = \alpha_h + \beta_h \varepsilon_t^{\ell} + \Gamma_h' X_{t-1} + u_{t+h}$$

where $GV_{t,t+h}$ represents the cumulative Growth minus Value differential and ε_t^{ℓ} represents the residual liquidity shock derived from the Eurosystem's reaction function.

This contribution should be interpreted with caution. The paper does not seek to demonstrate that the ECB directly causes a growth bubble or that European monetary policy necessarily results in capital misallocation. The objective is to assess whether the observed dynamic patterns are consistent with a financial duration channel. This approach preserves empirical rigor and avoids overinterpretation of an identification strategy that is, by design, reduced-form and conditional.

In summary, this paper addresses the gap at the intersection of European unconventional monetary policy, Eurosystem liquidity, intertemporal discounting, and relative asset valuation. The originality of the approach lies in examining the liquidity architecture not only as a mechanism for aggregate stabilization but also as a potential force for the relative realignment of asset prices with different financial durations.

5. Data and Variable Creation

5.1 Description of the dataset

The empirical analysis in this paper is based on a monthly macro-financial dataset specifically constructed to examine the relationships among Eurosystem liquidity, European financial conditions, and the relative valuation of growth and value assets. The observation unit is the month, and the primary empirical focus is the eurozone. However, some financial variables used as market proxies, particularly the Growth versus Value distinction, correspond to European indices and not strictly to the euro area. This distinction is explicitly noted throughout the chapter because it affects the geographic scope of the empirical interpretation.

The dataset combines five sets of information. The first set includes Eurosystem liquidity variables, primarily the aggregate balance sheet size and, when available, the banking system's excess liquidity. The second set incorporates financial conditions variables, such as the CISS systemic stress composite indicator and European implied volatility as approximated by VSTOXX. The third block includes monetary policy and term structure variables, such as the ECB's deposit facility rate, the €STR, and the two- and ten-year euro area sovereign yield curve rates. The fourth block incorporates aggregate stock market variables, primarily the EURO STOXX 50. The fifth block constructs the relative valuation variable "Growth minus Value" based on the MSCI Europe Growth and MSCI Europe Value indices.

The purpose of the dataset is not to provide an exhaustive description of all monetary transmission channels in the euro area, but rather to establish an empirical foundation consistent with the study's research question. In particular, the dataset should allow us to examine whether the ECB's and the Eurosystem's liquidity expansions, once initial financial conditions are taken into account, are associated with a relative improvement in assets with longer financial durations, approximated by Growth; compared to assets with shorter relative financial durations, approximated by Value.

The complete structure of the dataset, including its variable blocks, main variables, and empirical function within the study design, is presented in Appendix A.5. The main body of the text includes only the conceptual description necessary to understand the empirical strategy.

5.2 Frequency, time window, and aggregation criteria

The analysis is conducted on a monthly basis. There are three reasons for this choice. First, Eurosystem liquidity variables and relevant financial indicators are available at varying frequencies, including daily, weekly, and monthly, and in some cases, series with more limited availability. The monthly frequency allows these series to be harmonized without imposing excessively aggressive interpolations. The second reason, analysis using local projections requires a sufficiently broad time sample, but also a frequency compatible with the financial dynamics being studied. Finally, the third reason, the Growth versus Value section is constructed using monthly series downloaded manually; therefore, a higher frequency would not be consistent across all variables in the analysis.

The target time frame is defined as January 2000 through February 2026, subject to the actual availability of each series. In practice, the sample used in each specification may vary depending on the coverage of the included variables. For example, some ECB variables have long coverage, while others, such as €STR or excess liquidity in its direct form, may have shorter time windows. Therefore, this study distinguishes between the maximum time window of the dataset and the effective estimation

sample.

Daily or weekly data series are converted to monthly data using the latest available data point for the month. This rule is particularly suitable for asset prices, financial indices, interest rates, and stock variables, since the latest monthly value summarizes the information available at the end of the period. Formally, if x_τ denotes a daily or weekly observation within month t, the monthly observation is defined as:

$$x_t = x_{\max(\tau \in t)}$$

This convention applies to variables such as CISS, monetary policy rates, sovereign yield curve rates, Eurosystem total assets, excess liquidity, EURO STOXX 50, and VSTOXX, where applicable. For flow variables, if they were to be incorporated into future specifications, it might be more appropriate to use monthly sums or averages. However, the main specification of this study primarily uses stock variables, prices, indices, and rates, so the latest monthly data aggregation is the central convention.

The monthly time series is constructed using the EURO STOXX 50 as the preferred base series for time-series merging. The remaining variables are then integrated by monthly closing date. This decision does not imply that the EURO STOXX 50 is the study's main variable, but rather that it provides a broad and stable financial time series against which to align the rest of the data.

5.3 Eurosystem liquidity variables

Eurosystem liquidity is estimated using balance sheet variables and, when available, more direct measures of excess liquidity. The main variable is the Eurosystem's total assets, denoted by L_t. This measure broadly captures the expansion of the central bank's balance sheet and summarizes the combined effect of refinancing operations, asset purchases, and other assets matched by central bank liabilities.

Using total assets as a proxy for liquidity has a clear advantage: it provides a broad, observable measure with sufficient time coverage for monthly analysis. However, it also has a limitation. The size of the Eurosystem's balance sheet does not exactly correspond to the effective liquidity available to all segments of the financial system. It is a broad measure of monetary intervention and provision, not a microstructural measure of market liquidity.

The second liquidity variable considered is excess liquidity, generally defined as the liquidity held by the banking system in excess of minimum reserve requirements. In simplified terms, if R_t represents the reserves held by credit institutions and MR_t represents the required minimum reserves, excess liquidity can be expressed as:

$$EL_t = R_t - MR_t$$

This variable is conceptually closer to excess bank liquidity than total balance sheet size. However, its availability and temporal consistency may be more limited. For this reason, total Eurosystem assets remain the baseline measure, while excess liquidity can be used as a complement or to enhance robustness if the empirical coverage is sufficient.

Holdings under the Asset Purchase Programme (APP) and the Pandemic Emergency Purchase Programme (PEPP) are also considered as potential context or robustness variables. Conceptually, the APP and PEPP are relevant because they directly link balance sheet expansion to asset purchases

and to the transmission to interest rates, risk premiums, and financial fragmentation. However, they should only be incorporated into the models if the holdings series is correctly defined, downloaded, and time-aligned. In the absence of a complete operational series, the APP and PEPP should be treated as part of the institutional context and not as core variables in the estimated dataset.

The baseline liquidity variable is defined as follows:

$$\ell_t = \log(L_t)$$

given that $L_t > 0$. Its monthly growth is calculated as:

$$\Delta\ell_t = \ell_t - \ell_{t-1}$$

to express growth in approximate percentage points, the figure can be multiplied by 100:

$$g_t^L = 100(\ell_t - \ell_{t-1})$$

The logarithmic difference is preferable to the absolute change because it allows the variation to be interpreted as proportional growth and reduces dependence on the nominal scale of the balance sheet. This transformation is particularly suitable for positive, persistent, and upward-trending stock series, such as the size of the Eurosystem's balance sheet.

5.4 Financial Conditions Variables

Financial conditions are assessed using indicators of systemic stress, implied volatility, and, where available, credit spreads. The key variable in this section is the ECB-developed Composite Indicator of Systemic Stress (CISS). The CISS synthesizes tensions across different segments of the financial system and is particularly well-suited for the eurozone, where monetary transmission can be affected by episodes of fragmentation, banking stress, sovereign stress, and deteriorating market liquidity (Holló et al., 2012).

In this paper, the CISS serves two functions. First, it acts as a financial conditions variable within the Eurosystem's reaction function. If the central bank injects liquidity in response to episodes of stress, the CISS should help capture part of that endogenous reaction. Second, the CISS can be used as a state variable to define financial regimes. For example, a regime-dependent specification can define a high-stress indicator as:

$$D_t^{Stress} = \mathbf{1}\{CISS_t > Q_{0.75}(CISS)\}$$

where $Q_{0.75}(CISS)$ represents the 75th percentile of the sample distribution of the indicator. This definition makes it possible to distinguish between periods of financial stability and periods of heightened stress.

The second financial condition variable is VSTOXX, an implied volatility index linked to the EURO STOXX 50. While CISS captures broad systemic stress, VSTOXX more directly measures the expected volatility of the European stock market. Its inclusion allows us to distinguish between aggregate systemic stress and equity-specific volatility.

Depending on the specification, CISS and VSTOXX can be used as levels or changes. For the reaction function, the level of financial stress is economically meaningful, since a central bank responds

to the current financial conditions. For local projections, when the dependent variable is persistent, a cumulative change between $t-1$ and $t+h$ can be used. In the case of the CISS:

$$\Delta_h CISS_{t+h} = CISS_{t+h} - CISS_{t-1}$$

If VSTOXX is used in logarithms, the following is defined:

$$v_t = \log(VSTOXX_t)$$

and its monthly change as:

$$\Delta v_t = v_t - v_{t-1}$$

The choice between levels, changes, or cumulative changes should be consistent with the economic interpretation of each specification. In general, the level of stress is useful for modeling the liquidity response, while the subsequent change in stress or volatility is useful for studying the dynamic transmission.

5.5 Monetary Policy Variables and the Term Structure of Interest Rates

The monetary policy and interest rate term structure block incorporates variables that approximate the risk-free component of the discount rate and the ECB's monetary policy stance. The first variable is the deposit facility rate, i_t^{DFR}, which has taken on central importance in the ECB's operational framework, especially during periods of reserve abundance. The deposit facility acts as an anchor for very short-term monetary rates and directly reflects the stance of conventional monetary policy.

The second variable is the €STR, the euro short-term interest rate. Although its time series is more recent than that of other variables, it is useful as a complementary indicator of the overnight money market. Due to this time-series limitation, the €STR can be used in descriptive analyses or as a robustness check, but not necessarily in all baseline specifications.

The term structure of interest rates is approximated using the two-year and ten-year euro area sovereign yields, denoted by i_t^{2Y} and i_t^{10Y}. These yields allow us to construct the slope of the curve:

$$slope_t = i_t^{10Y} - i_t^{2Y}$$

Its monthly variation is defined as:

$$\Delta slope_t = slope_t - slope_{t-1}$$

The slope of the yield curve encapsulates information about future interest rate expectations, term premiums, the economic cycle, and financial conditions. In the context of this study, its significance lies in the fact that it links the Eurosystem's actions to the discount channel. If liquidity injections and asset purchase programs suppress long-term yields or alter term premiums, the term structure of interest rates should reflect part of that mechanism.

For interest rates and spreads expressed as percentages, the baseline transformation is the simple monthly change:

$$\Delta i_t = i_t - i_{t-1}$$

This transformation is preferable to the logarithmic transformation because interest rates can be negative, as was the case in the eurozone during part of the sample period. Consequently, no logarithmic transformation is applied to monetary policy rates, money market rates, or sovereign bond yields.

5.6 Aggregate Stock Market Variables

The aggregate stock market is represented by the EURO STOXX 50. This index serves as a benchmark for European equities and captures the market's overall response to changes in liquidity, financial stress, interest rates, and monetary conditions. Although the primary focus of this study is on Growth versus Value, the EURO STOXX 50 is necessary for monitoring and describing the aggregate stock market.

Let P_t^{SX5E} be the monthly level of the index. Its logarithm is defined as:

$$p_t^{SX5E} = \log(P_t^{SX5E})$$

and its monthly return as:

$$r_t^{SX5E} = p_t^{SX5E} - p_{t-1}^{SX5E}$$

When expressed in approximate percentage points:

$$R_t^{SX5E} = 100(p_t^{SX5E} - p_{t-1}^{SX5E})$$

The return on the EURO STOXX 50 can be used in three ways. First, as a dependent variable to study the aggregate effect of liquidity innovations on the stock market. Second, as a control in the Eurosystem's reaction function, given that stock market declines may be associated with financial distress and policy responses. Third, as a comparison variable between the Growth and Value sectors, allowing us to distinguish between aggregate market appreciation and relative reallocation among investment styles.

It is important to emphasize that the EURO STOXX 50 does not replace the Growth versus Value framework. The aggregate index measures the overall performance of large European companies, while the Growth minus Value variable captures a cross-sectional dimension of relative valuation. The central hypothesis of this study is not limited to the idea that liquidity raises the aggregate price of equities, but rather that it may favor, relatively speaking, assets with longer financial durations.

5.7 The "Growth minus Value" Model: Transformations and Proxy Limitations

The MSCI Europe Growth and MSCI Europe Value series used in this study were obtained from Curvo. The series were manually downloaded at monthly frequency, are denominated in euros, and contain a date variable and a performance-level variable for each index. Because the econometric analysis is based on returns rather than index levels, the monthly logarithmic return of any positive index P_t is defined as

$$r_t = \log(P_t) - \log(P_{t-1}).$$

Applying this transformation separately to the two portfolios, the Growth minus Value differential is defined as

$$GV_t = r_t^{\text{Growth}} - r_t^{\text{Value}}.$$

Under this convention, $GV_t > 0$ indicates that Growth outperforms Value during month t, whereas $GV_t < 0$ indicates relative outperformance of Value. For the cumulative dependent variable used in the local projections, the differential over horizon h is constructed as

$$GV_{t,t+h} = \sum_{j=0}^{h} \left(r_{t+j}^{\text{Growth}} - r_{t+j}^{\text{Value}} \right).$$

The remaining transformations, aggregation criteria, and sign conventions are reported in Appendix A.3. The principal limitation of this framework is geographical. Although the primary institutional focus of the study is the euro area, the available operational series correspond to MSCI Europe Growth and MSCI Europe Value rather than MSCI EMU Growth and MSCI EMU Value. MSCI Europe covers developed European markets and therefore includes both euro-area and non-euro-area countries. Consequently, GV_t should be interpreted as a European proxy for the Growth versus Value dimension, rather than as a strictly euro-area measure of relative valuation. The remaining limitations of this proxy, including differences in sectoral composition, factor exposure, country coverage, and currency exposure, are discussed in Section 8.6.

6. Econometric Methodology and Identification

This chapter outlines the econometric strategy for analyzing the dynamic relationships among ECB and Eurosystem liquidity, European financial conditions, and the relative valuation of assets with different durations. The central difficulty is that central bank liquidity cannot be treated as an exogenous variable: balance sheet expansions, funding operations, and purchase programs tend to intensify precisely during episodes of stress, high volatility, widening risk premiums, or a deterioration in monetary transmission.

For this reason, the empirical design is organized into four sections. First, we estimate a reaction function for Eurosystem liquidity to separate the systematic component of the response to adverse financial conditions from a residual innovation; second, local projections are used to estimate dynamic responses by horizon, following the methodology of Jordà (2005); third, a regime-dependent specification is introduced to assess whether transmission differs during periods of financial stress; and fourth, the Growth versus Value block is estimated, which links the econometric methodology to the financial duration hypothesis.

The interpretation of the results should be framed in terms of conditional dynamic evidence. The strategy does not identify pure structural monetary shocks in the sense of high-frequency or exogenous identification, but rather residual liquidity innovations conditional on the set of information incorporated into the reaction function. This clarification is essential to avoid an overly strong causal interpretation.

6.1 Endogeneity of Liquidity and the Reaction Function

Let L_t be a measure of Eurosystem liquidity, such as total balance sheet assets or excess liquidity, and let $\ell_t = \log L_t$ be its logarithmic transformation. The monthly change in liquidity is defined as

$$\Delta \ell_t = 100(\ell_t - \ell_{t-1}) \tag{1}$$

A naive specification that directly links $\Delta \ell_t$ to a financial variable y_{t+h} would not allow us to distinguish between the transmission effect and the central bank's endogenous reaction. If the Eurosystem expands liquidity when systemic stress, volatility, or financial fragmentation increase, a contemporaneous correlation between liquidity and financial deterioration may reflect the monetary authority's stabilizing response rather than an adverse effect of liquidity.

To mitigate this problem, a liquidity reaction function is estimated:

$$\Delta \ell_t = \alpha + \sum_{i=1}^{p} \rho_i \Delta \ell_{t-i} + \sum_{i=0}^{q} \gamma_i' S_{t-i} + \sum_{i=0}^{m} \delta_i' Z_{t-i} + \varepsilon_t^{\ell} \tag{2}$$

The vector S_t captures financial conditions, including CISS, VSTOXX, aggregate stock market returns, and stress indicators. The vector Z_t incorporates additional macrofinancial controls, such as the deposit facility rate, the yield curve slope, or changes in market interest rates. The term ε_t^{ℓ} represents the residual liquidity innovation, that is, the portion of the change in liquidity that is not explained by past liquidity dynamics or by the financial conditions included in the reaction function.

This first step does not treat liquidity as a perfectly exogenous shock. Its role is more modest but methodologically significant: to eliminate the predictable portion of the Eurosystem's response and use the residual component as a variable of interest for subsequent dynamic analysis.

6.2 Residual Liquidity Innovations

Based on the estimation of equation (2), the estimated residual is constructed as

$$\widehat{\varepsilon}_t^{\ell} = \Delta \ell_t - \widehat{\Delta \ell}_t \tag{3}$$

To make it easier to interpret the values, the innovation can be standardized by dividing it by its sample standard deviation:

$$\widetilde{\varepsilon}_t^{\ell} = \frac{\widehat{\varepsilon}_t^{\ell}}{\widehat{\sigma}_{\varepsilon^{\ell}}} \tag{4}$$

With this normalization, the coefficient estimated in the local projections measures the response of the dependent variable to a residual liquidity shock equivalent to one standard deviation. This convention improves comparability across specifications and prevents the interpretation of coefficients that depend on the liquidity variable's original scale.

Residual innovation is interpreted as conditional innovation, not as a structural monetary shock. It may continue to incorporate components of expectations, unobserved policy decisions, announcements, changes in collateral, or institutional factors not included in S_t and Z_t. Therefore, the empirical framework of this study is formulated in terms of conditional dynamic responses and evidence consistent with theoretical channels.

6.3 Local Projections: Baseline Specifications and Cumulative Responses

The second stage estimates dynamic responses using local projections. For each horizon $h = 0, 1, \ldots, H$, a separate regression is estimated of the form

$$Y_{t,h} = \alpha_h + \beta_h \widetilde{\varepsilon}_t^{\ell} + \Gamma_h' X_{t-1} + u_{t+h}^{(h)} \tag{5}$$

The variable $Y_{t,h}$ is defined according to the economic nature of the dependent variable. For level variables or persistent indicators, such as CISS or VSTOXX, a cumulative change relative to the period prior to the innovation is used:

$$Y_{t,h} = y_{t+h} - y_{t-1} \tag{6}$$

For stock market returns or return differentials, a cumulative return is used:

$$R_{t,t+h} = \sum_{j=0}^{h} r_{t+j} \tag{7}$$

The vector X_{t-1} includes lagged controls to prevent the coefficient β_h from mechanically capturing initial financial conditions. According to the specification, these controls include CISS, VSTOXX, aggregate stock returns, the monetary policy stance, the yield curve slope, and the lags of the dependent variable. The use of lagged variables preserves the dynamic interpretation of the innovation at t and reduces the risk of contemporary simultaneity.

The sequence $\{\widehat{\beta}_h\}_{h=0}^{H}$ constitutes the estimated response trajectory. A negative value of $\widehat{\beta}_h$ for CISS or VSTOXX would be consistent with a reduction in financial stress or volatility. A positive value for aggregate returns would indicate a cumulative improvement in the stock market. In the Growth

versus Value section, the expected sign is positive if the financial duration channel dominates.

6.4 Regime Dependence

The transmission of liquidity may depend on the initial financial position. A given liquidity innovation does not necessarily have the same effect in a normal environment as it does during a period of systemic stress, balance sheet constraints, high risk aversion, or financial fragmentation. To capture this heterogeneity, a regime variable is defined:

$$D_t^{Stress} = \mathbf{1}\{CISS_t > Q_{0.75}(CISS)\} \tag{8}$$

The specification with interaction is

$$Y_{t,h} = \alpha_h + \beta_h \widetilde{\varepsilon}_t^{\ell} + \theta_h \left(\widetilde{\varepsilon}_t^{\ell} \times D_t^{Stress}\right) + \lambda_h D_t^{Stress} + \Gamma_h' X_{t-1} + u_{t+h}^{(h)} \tag{9}$$

In this specification, β_h measures the estimated response under normal conditions, and $\beta_h + \theta_h$ measures the total response under stress. If liquidity is more effective when markets are under stress, a larger response should be observed under stress, particularly in variables such as CISS, VSTOXX, aggregate returns, or valuation spreads.

The baseline scenario is defined using CISS because this indicator captures systemic stress across various segments of the European financial system. For robustness, alternative thresholds, such as $Q_{0.80}$ or $Q_{0.90}$, or a definition based on VSTOXX, may be used.

6.5 Growth versus Value

The main section of the paper examines whether liquidity innovations are associated with a relative outperformance of growth stocks relative to value stocks. The main variable is defined as

$$GV_t = r_t^G - r_t^V \tag{10}$$

where r_t^G represents the return on the Growth portfolio and r_t^V the return on the Value portfolio. Under this convention, $GV_t > 0$ indicates that Growth outperforms Value in period t. To study the transmission across different horizons, the cumulative differential is used:

$$GV_{t,t+h} = \sum_{j=0}^{h} \left(r_{t+j}^G - r_{t+j}^V\right) \tag{11}$$

The baseline specification for the block is

$$GV_{t,t+h} = \alpha_h + \beta_h \widetilde{\varepsilon}_t^{\ell} + \Gamma_h' X_{t-1} + u_{t+h}^{(h)} \tag{12}$$

The central empirical hypothesis is

$$H_0 : \beta_h = 0, \qquad H_1 : \beta_h > 0 \quad \text{for a relevant set of horizons } h \tag{13}$$

The economic rationale stems from the concept of financial duration. If Growth concentrates a larger proportion of its value in distant expected cash flows, and Value relies relatively more on current earnings, existing assets, or near-term cash flows, Growth should be more sensitive to a compression

in the discount rate. In short, for a portfolio $i \in \{G, V\}$,

$$\Delta \ln P_t^i \approx -D_t^i \Delta R_t^i \tag{14}$$

If $D_t^G > D_t^V$ and a liquidity shock reduces the common component of the discount, then

$$\Delta \ln P_t^G - \Delta \ln P_t^V \approx -(D_t^G - D_t^V)\Delta R_t > 0 \quad \text{if } \Delta R_t < 0 \tag{15}$$

This prediction does not imply that GV_t is a pure measure of financial duration. Growth and Value also differ in terms of sector composition, country, currency, market capitalization, factor exposure, and cyclical sensitivity. Furthermore, in the absence of operational series for MSCI EMU Growth and MSCI EMU Value, the study uses MSCI Europe Growth and MSCI Europe Value as available European proxies. Therefore, the correct interpretation is that the Growth versus Value comparison provides European evidence consistent with a financial duration channel, not a strictly pure measure of relative valuation within the eurozone.

6.6 Inference, Robustness, and Scope of Identification

Inference in local projections must account for the autocorrelation induced by temporal overlap, especially when the dependent variable is defined as a cumulative response. To this end, standard errors robust to heteroscedasticity and autocorrelation (HAC) are used, following the logic of Newey and West (1987). For each horizon h, the inference regarding $\widehat{\beta}_h$ is based on

$$t_h = \frac{\widehat{\beta}_h}{\widehat{se}_{HAC}(\widehat{\beta}_h)} \tag{16}$$

The HAC baseline window is set to

$$b_h = h + 1 \tag{17}$$

so that it increases with the accumulation horizon. This adjustment prevents confidence intervals from being artificially narrow over long horizons. The literature on multi-period returns and overlapping data has a classic precedent in Hansen and Hodrick (1980).

The robustness of the design is assessed by varying the liquidity measure, using alternative controls, applying different stress thresholds, and specifying models with and without extreme events, and by using alternative definitions of the “Growth minus Value” variable when the data permit. The objective of these tests is not to obtain a single definitive estimate, but rather to verify whether the sign, the time profile, and the economic interpretation of the responses hold up under reasonable changes in the specification.

The main limitation of the design is that the residual liquidity innovation does not amount to a pure exogenous shock. The reaction function reduces the observed endogeneity, but does not completely eliminate bias due to omitted variables, simultaneity, or changes in expectations. A stronger identification would require external instruments, high-frequency identification around monetary policy announcements, structural constraints, or a narrative design. In the absence of such a strategy, the paper adopts a reduced-form interpretation: the results are taken to reflect conditional dynamic responses consistent with the theoretical channels, not definitive structural causality.

7. Empirical Results

This chapter presents the empirical results of the study. The discussion follows the identification framework outlined in the methodology chapter. First, we analyze the data availability and the descriptive structure of the monthly panel. Second, we examine the time-series properties of the data, with particular attention to the stationarity of levels, returns, changes, and growth rates. Third, we present the estimation of the Eurosystem's liquidity reaction function, which aims to separate the systematic component of the monetary policy response from residual liquidity innovation. Next, we estimate the dynamic responses of various financial variables using local projections: first, financial conditions; then, aggregate stock market returns; and subsequently, in regime-dependent specifications. Finally, the central section of the paper is analyzed: the relative response of Growth versus Value, interpreted as an empirical approximation of the differential sensitivity of assets with different financial durations.

The results should be interpreted with caution. The estimates do not identify a fully exogenous structural monetary shock; rather, they capture dynamic responses conditional on a residual liquidity innovation, after controlling for the Eurosystem's reaction function. Therefore, the appropriate empirical language is that of conditional association, estimated dynamic response, and evidence consistent with the theoretical channels developed in previous chapters. This caution is particularly important because central bank liquidity tends to expand in contexts of financial distress, high volatility, and market stress, which can generate seemingly counterintuitive contemporaneous signals.

7.1 Descriptive Analysis and Data Availability

The main empirical dataset is compiled on a monthly basis and covers the period from January 2000 to February 2026. The sample contains 314 monthly observations, allowing for a joint analysis of several key phases in European monetary policy: the period preceding the global financial crisis, the sovereign debt crisis, the era of negative interest rates, the expansion of the Eurosystem's balance sheet, the pandemic, and the subsequent monetary normalization.

The core variables in the analysis provide sufficient coverage to support the main empirical design. In particular, the aggregate stock market index *EURO STOXX 50*, the CISS composite indicator of systemic stress, the deposit facility rate, and the total assets of the Eurosystem are available throughout the monthly sample, with no missing values. The associated dynamic transformations, such as the monthly return on the stock index, the monthly change in the CISS, the change in the deposit rate, and the logarithmic growth of the Eurosystem's total assets, exhibit the usual initial missing value resulting from time-series differentiation.

Table 1

Availability of the main variables in the monthly dataset

Variable	First valid date	Last valid date	Valid observations	Missing	Empirical use
EURO STOXX 50	2000-01-31	2026-02-28	314	0	Market level; basis for returns.
EURO STOXX 50 Return	2000-02-29	2026-02-28	313	1	Aggregate equity-market response and reaction-function control.
CISS	2000-01-31	2026-02-28	314	0	Financial Stress and Regime Definition.
Deposit facility rate	2000-01-31	2026-02-28	314	0	Monetary policy control.
Total assets of the Eurosystem	2000-01-31	2026-02-28	314	0	Basis for measuring balance sheet expansion.

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Variable	First valid date	Last valid date	Valid observations	Missing	Empirical use
Growth in Eurosystem assets	2000-02-29	2026-02-28	313	1	Baseline liquidity-change measure.
10-Year–2-Year Spread in the Eurozone	2004-09-30	2026-02-28	258	56	Time-series analysis.
Excess liquidity	2024-09-30	2026-02-28	18	296	Conceptual proxy; not a full-sample baseline.
VSTOXX	–	–	0	314	Conceptual proxy; not a full-sample baseline. Pending due to lack of usable data series.
APP holdings	–	–	0	314	Pending; not used in the central estimate.
PEPP holdings	–	–	0	314	Pending; not used in the central estimate.
MSCI Europe Growth	2000-01-31	2026-02-28	314	0	Developed-European Growth index used to construct portfolio returns.
MSCI Europe Value	2000-01-31	2026-02-28	314	0	Developed-European Value index used to construct portfolio returns.
Growth minus Value	2000-01-31	2026-02-28	314	0	Dependent variable for the Growth minus Value model.

Note. The Growth and Value variables are derived from the MSCI Europe Growth and MSCI Europe Value indices. They should therefore not be interpreted as pure MSCI EMU portfolios.

Source: Own elaboration.

This liquidity structure justifies using the growth of the Eurosystem's total assets as the benchmark liquidity measure. Formally, if L_t represents the Eurosystem's total assets, it is defined as:

$$\ell_t = \log L_t$$

and the monthly growth in liquidity is estimated using:

$$\Delta \ell_t = \ell_t - \ell_{t-1}$$

In some tables and specifications, this variation is scaled or standardized to facilitate the interpretation of the coefficients. The choice of total Eurosystem assets as the baseline proxy does not imply that it is a perfect measure of liquidity available to the financial system. Its main advantage is its complete time coverage and its interpretation as a broad measure of the central bank's balance sheet size. In contrast, the direct excess liquidity series has very limited time coverage, available only from September 2024, and therefore cannot serve as a baseline variable for the entire sample.

The CISS variable provides the primary measure of systemic financial stress. Its full monthly availability allows it to be used as a dependent variable, a control variable, and a regime criterion. From a descriptive standpoint, CISS has a mean of 0.1589 and a standard deviation of 0.1855, reflecting the presence of episodes of financial stress concentrated in specific periods of the sample. The monthly return of the *EURO STOXX 50* has a mean of 0.0008 and a standard deviation of 0.0462, consistent with a stock market return series of relatively high volatility. The monthly growth of the Eurosystem's total assets has a mean of 0.0067 and a standard deviation of 0.0306, reflecting balance sheet dynamics characterized by intense expansions during certain episodes, as well as phases of stabilization and normalization.

The euro area yield curve, constructed using two- and ten-year yields, has been available since September 2004. Therefore, specifications that incorporate the curve's slope use a shorter effective sample than the minimal specifications based on liquidity, CISS, stock market returns, and the deposit rate. This restriction does not invalidate the analysis, but it requires interpreting the models with curve

slope as extended specifications rather than as the study's sole empirical core.

The Growth versus Value panel is constructed using the MSCI Europe Growth and MSCI Europe Value indices. The raw data series cover a broad time window, from December 1974 to April 2026, with 617 aligned monthly observations and 616 aligned monthly returns. However, to maintain consistency with the rest of the European panel, the harmonized analysis is restricted to the common sample from 2000M1 to 2026M2. The central variable is defined as:

$$GV_t = r_t^{Growth} - r_t^{Value}$$

Under this convention, $GV_t > 0$ indicates that Growth outperforms Value in period t. This definition is consistent with the central hypothesis of this study: if liquidity compresses discount rates or risk premiums, and if Growth has a longer financial duration than Value, then the expected response of GV_t to an expansionary liquidity innovation should be positive over the relevant time horizons.

As noted in Section 5.7 and discussed in Section 8.6, the MSCI Europe indices represent developed Europe rather than the euro area, so this block provides European evidence consistent with the duration channel rather than a strictly euro-area measure.

7.2 Stationarity and Temporal Properties of Time Series

Before estimating the reaction function and local projections, the time-series properties were analyzed using ADF and KPSS tests. These tests are not used as the sole mechanical criterion, but rather as a guide for selecting transformations consistent with the economic nature of the variables. The combination of ADF and KPSS allows us to determine whether the time series exhibits behavior consistent with stationarity in levels or, conversely, requires differences, returns, or growth rates.

Table 2
ADF and KPSS tests for stationarity

Variable	ADF p-value	KPSS p-value	Decision
log EURO STOXX 50	0.262	0.016	Non-stationary
EURO STOXX 50 Return	< 0.001	0.100	Stationary
CISS	0.002	0.100	Stationary
Δ CISS	< 0.001	0.100	Stationary
Deposit facility rate	0.108	0.023	Non-stationary
Δ deposit facility rate	< 0.001	0.100	Stationary
log Eurosystem total assets	0.776	0.010	Non-stationary
Growth in Eurosystem assets	< 0.001	0.100	Stationary
Euro-area 10-year–2-year yield spread	0.353	0.031	Non-stationary
Change in the euro-area 10-year–2-year yield spread	< 0.001	0.100	Stationary
MSCI Europe Growth return	< 0.001	0.090	Stationary
MSCI Europe Value return	< 0.001	0.100	Stationary
Growth minus Value	< 0.001	0.100	Stationary

Note. The stationarity decision combines the evidence from the ADF and KPSS tests. KPSS p-values equal to 0.100 correspond to the upper reporting limit of the test.

Source: Own elaboration.

The results show a clear distinction between persistent levels and dynamic changes. The logarithm of the *EURO STOXX 50* shows evidence of nonstationarity, with an ADF *p*-value of 0.2624 and a KPSS *p*-value of 0.0162. In contrast, the monthly return of the index is stationary, with an ADF *p*-value of 0.0000 and a KPSS *p*-value of 0.1000. This pattern is consistent with the financial time series

literature: stock price levels exhibit high persistence, while returns are stationary.

The CISS indicator shows evidence consistent with stationarity at the level. The ADF test yields a *p*-value of 0.0017, and the KPSS test yields a *p*-value of 0.1000. The monthly variation of the CISS also appears to be stationary, with an ADF *p*-value of 0.0000 and a KPSS *p*-value of 0.1000. This evidence allows the CISS to be used at both levels, as a measure of financial stress, and in cumulative form, depending on the dynamic response specification.

The deposit facility rate exhibits more persistent behavior at the level. The ADF *p*-value of 0.1083 and the KPSS *p*-value of 0.0229 suggest nonstationarity at the level of the monetary policy rate. However, its monthly change is stationary, with an ADF *p*-value of 0.0000 and a KPSS *p*-value of 0.1000. Therefore, dynamic specifications preferentially use Δi_t when the objective is to model changes in monetary conditions.

The liquidity variable exhibits the expected pattern. The logarithm of the Eurosystem's total assets is non-stationary, with an ADF *p*-value of 0.7757 and a KPSS *p*-value of 0.0100. In contrast, the monthly growth of total assets is stationary, with an ADF *p*-value of 0.0000 and a KPSS *p*-value of 0.1000. This result justifies estimating the reaction function not on the level of the balance sheet, but on its rate of change:

$$\Delta \ell_t = \ell_t - \ell_{t-1}$$

The slope of the euro area curve, defined as the difference between the 10-year and 2-year yields, also shows evidence of non-stationarity in levels. The ADF *p*-value is 0.3530 and the KPSS *p*-value is 0.0306. Its monthly variation, however, is stationary, with an ADF *p*-value of 0.0000 and a KPSS *p*-value of 0.1000. Therefore, the change in slope is preferred in the reaction function and in the local projections.

In the Growth versus Value section, the levels of the MSCI Europe Growth and MSCI Europe Value indices exhibit non-stationarity, whereas returns and return spreads are stationary. This result is particularly important because it validates the use of GV_t as a dependent variable in dynamic models. Since the variable GV_t is defined as the difference in returns, it avoids the need to deal with persistently rising index levels and is interpreted directly as the relative performance of Growth versus Value.

Overall, the stationarity tests support the adopted transformation strategy. Prices and stock market indices are modeled using returns; liquidity using logarithmic balance sheet growth; interest rates using monthly changes; the slope using variations; and Growth versus Value using return differentials. This structure reduces the risk of spurious regressions and preserves the economic interpretation of the variables.

7.3 Results of the Eurosystem's Reaction Function

The first empirical stage consists of estimating a reaction function for Eurosystem liquidity. The objective is not to provide an exhaustive explanation of all ECB decisions, but rather to obtain a residual liquidity innovation that accounts, at least partially, for the systematic and predictable component of balance sheet expansion in response to observable financial conditions. This stage is fundamental because central bank liquidity is not exogenous: it tends to increase precisely during episodes of stress, stock market declines, or financial turmoil.

The general specification can be written as:

$$\Delta \ell_t = \alpha + \sum_{i=1}^{p} \rho_i \Delta \ell_{t-i} + \sum_{i=1}^{p} \gamma_i CISS_{t-i} + \sum_{i=1}^{p} \delta_i r^M_{t-i} + \sum_{i=1}^{p} \phi_i \Delta i_{t-i} + \sum_{i=1}^{p} \psi_i \Delta slope_{t-i} + \varepsilon^{\ell}_t$$

where $\Delta \ell_t$ represents the monthly growth in the Eurosystem's total assets, $CISS_t$ measures systemic financial stress, r^M_t represents the monthly return on the *EURO STOXX 50*, Δi_t is the monthly change in the deposit facility rate, and $\Delta slope_t$ is the monthly change in the slope of the 10Y minus 2Y curve. The term ε^{ℓ}_t is the residual liquidity innovation subsequently used in the local projections.

The preferred model uses three lags, $p = 3$, and an effective sample size of 254 observations. The choice of this specification is based on information criteria and residual diagnostics. The model has an AIC of -1064.9970, a BIC of -1008.3997, an adjusted R^2 of 0.1494, and a Durbin-Watson statistic of 1.9539. The Ljung-Box tests do not reject residual autocorrelation at lags 6 and 12, with *p*-values of 0.1807 and 0.2367, respectively. In contrast, the Jarque-Bera and Breusch-Pagan tests suggest non-normality and residual heteroscedasticity, with *p*-values of 0.0000 and 0.0005. This evidence justifies the subsequent use of robust standard errors and a cautious interpretation of the residual innovation.

The estimated reaction function ($p = 3$, $N = 254$, adjusted $R^2 = 0.149$) confirms that Eurosystem liquidity contains a systematic, predictable component. Balance-sheet growth is highly persistent, with the third lag entering significantly (0.2582, $p < 0.001$); while the third lag of the EURO STOXX 50 return enters negatively (-0.1323, $p = 0.036$), consistent with a stabilizing reaction in which prior equity declines precede subsequent balance-sheet expansion. The third lag of the change in the deposit rate is also negative and weakly significant (-0.0315, $p = 0.075$). The full coefficient table and residual diagnostics are reported in Table 6 of Appendix A.6.1.

Figure 5 in Appendix A.6.1 plots observed liquidity growth against the predictable component implied by the reaction function, illustrating that the systematic response captures the broad expansion phases while leaving substantial residual variation.

The key insight of this stage lies not in each individual coefficient but in the construction of a residual liquidity innovation. This innovation is defined as:

$$\widehat{\varepsilon}^{\ell}_t = \Delta \ell_t - \widehat{\Delta \ell_t}$$

In the specifications that follow, the standardized version, *liquidity_innovation_std*, is typically used to make the coefficients easier to read. One unit of this variable represents a residual liquidity innovation equivalent to one standard deviation. This transformation enables comparison of dynamic responses across variables expressed on different scales.

It is important to emphasize that $\widehat{\varepsilon}^{\ell}_t$ should not be interpreted as a pure structural monetary shock. The reaction function controls for a limited set of observable variables, but does not eliminate all possible endogeneity. Therefore, the residual innovation should be understood as a disciplined proxy for variation not explained by the systematic reaction included in the model. Its utility lies in improving identification beyond a simple correlation between liquidity and asset prices, not in providing definitive causal identification.

7.4 Dynamic Effects on Financial Conditions and Aggregate Returns

Once the residual liquidity innovation has been obtained, local projections are estimated to analyze its dynamic relationship with aggregate financial variables. The general specification is:

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h \hat{\varepsilon}_t^{\ell} + \Gamma_h' X_{t-1} + u_{t+h} \qquad h = 0, 1, \ldots, 24$$

where y_{t+h} represents the dependent variable at horizon h, $\hat{\varepsilon}_t^{\ell}$ is the standardized residual liquidity innovation, and X_{t-1} includes lagged controls. Estimation by horizons allows us to distinguish between contemporaneous associations and cumulative responses over the medium term.

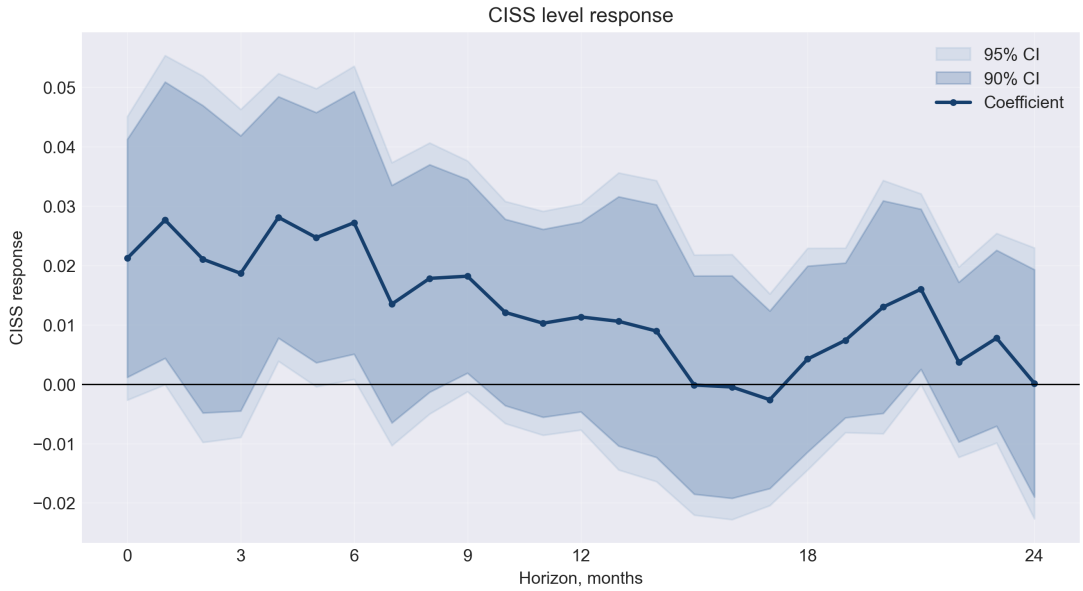


Figure 1
Estimated response of the CISS to a residual liquidity innovation

Source: Author's own calculations and local-projection estimates based on ECB Data Portal data (European Central Bank, 2026b).

The first set of results focuses on CISS, which is used as a measure of systemic financial stress. For the response in CISS levels, the coefficients show a positive association over some horizons. At $h = 0$, the estimated coefficient is 0.0212, with a 90% confidence interval of $[0.0012, 0.0412]$ and a p-value of 0.0831. At $h = 6$, the coefficient rises to 0.0272, with a 90% confidence interval of $[0.0051, 0.0493]$ and a p-value of 0.0444. In contrast, statistical precision disappears at longer horizons.

This result should be interpreted with caution. A positive response of the CISS to a residual liquidity innovation does not imply that liquidity causes financial stress. Given the endogenous nature of central bank action, it is plausible that even the residual innovation retains information about episodes of stress that the controls do not fully capture. In this sense, the result is consistent with a "liquidity as reaction" interpretation: liquidity appears associated with stress conditions because the Eurosystem intervenes more intensively in adverse financial environments.

Monetary and yield curve variables show a weaker signal. For the change in the deposit rate, the

selected coefficients do not exhibit clear statistical significance across the main horizons. In the case of the 10Y minus 2Y yield curve slope, the most visible result appears at the contemporaneous horizon, with a coefficient of 0.0315 at $h = 0$, a 90% confidence interval of $[0.0046, 0.0584]$, and a *p*-value of 0.0548. In later horizons, the statistical significance disappears. This pattern points to a contemporary association between liquidity innovations and yield curve movements, but does not allow us to assert a persistent transmission across the term structure of interest rates.

The second section analyzes aggregate stock market returns, as approximated by the *EURO STOXX 50*. In this case, the main dependent variable is the cumulative return between t and $t + h$, defined as:

$$R^{M}_{t,t+h} = \sum_{j=0}^{h} r^{M}_{t+j}$$

where r^{M}_{t} represents the monthly return of the aggregate stock index. This transformation allows us to capture dynamic effects that may not fully materialize in the current month.

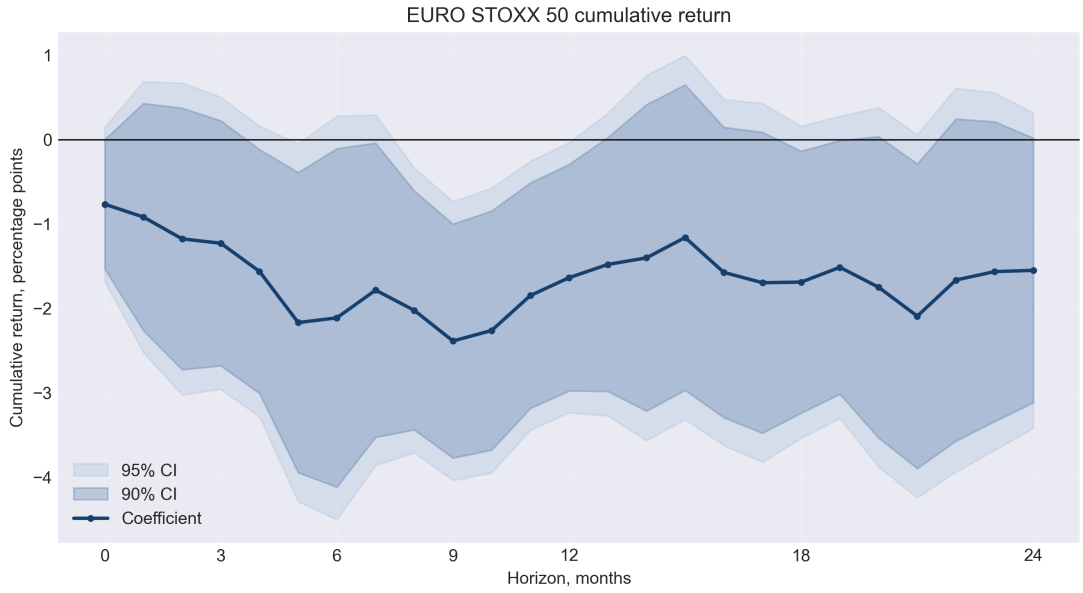


Figure 2
The EURO STOXX 50's cumulative response to a residual liquidity innovation

Source: Author's own calculations and local-projection estimates based on ECB Data Portal data (European Central Bank, 2026b).

The results show a negative conditional association between residual liquidity innovation and the cumulative returns of the *EURO STOXX 50* over various time horizons. At $h = 0$, the estimated coefficient is -0.7667 percentage points, with a 90% confidence interval of $[-1.5331, -0.0004]$. At $h = 6$, the coefficient is -2.1127, with a 90% confidence interval of $[-4.1183, -0.1071]$. At $h = 12$, the coefficient is -1.6370, with a 90% confidence interval of $[-2.9793, -0.2947]$. At $h = 24$, the estimated coefficient is -1.5504, although the 90% confidence interval $[-3.1185, 0.0176]$ marginally includes zero.

At first glance, this result might seem to contradict the hypothesis that liquidity stabilizes financial

markets. However, a correct interpretation must account for the endogeneity of the central bank's reaction function. Residual liquidity shocks, even after partial stripping of the systematic reaction included in the model, may still be associated with unobserved or incompletely captured episodes of financial distress. In that case, a negative response in stock returns does not necessarily imply that liquidity causally reduces asset prices. Rather, it may indicate that unexpected liquidity increases occur in contexts where the market is already deteriorating or anticipating adverse conditions.

Overall, local projections of financial conditions and aggregate returns provide mixed evidence. CISS responds positively over some horizons, reinforcing the importance of endogeneity and the reaction function. Monetary policy and slope variables exhibit weaker and less persistent effects. The cumulative returns of the *EURO STOXX 50* show a negative response across several horizons, consistent with an interpretation of intervention in times of stress. Therefore, the aggregate results do not constitute the clearest channel for testing the financial duration hypothesis. The Growth versus Value section becomes more important because it allows analysis of a relative dimension closer to differential discount sensitivity.

7.5 Regime Dependence: Transmission During Periods of Stress

The methodology section noted that liquidity transmission may depend on the financial regime. To assess this possibility, a high-stress dummy variable is defined based on the lagged CISS. The general specification is:

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h \widehat{\varepsilon}_t^{\ell} + \theta_h \left(\widehat{\varepsilon}_t^{\ell} \times D_t^{Stress} \right) + \lambda_h D_t^{Stress} + \Gamma_h' X_{t-1} + u_{t+h}$$

In this formulation, β_h measures the estimated response under normal conditions, and $\beta_h + \theta_h$ measures the total response under stress conditions. The dummy variable D_t^{Stress} is constructed using percentiles of $CISS_{t-1}$. In the baseline specification, the p75 threshold corresponds to a lagged CISS value of 0.2335. Under this criterion, the sample contains 78 high-stress observations and 235 normal-regime observations, with a stress proportion of 24.92%.

The regime-dependent estimates display heterogeneity across horizons but no monotonic or robust amplification under stress. For the EURO STOXX 50, the stress response is markedly more negative at $h = 6$, whereas the ordering of the regime-specific responses reverses at $h = 12$; the CISS responses are similarly unstable in sign across horizons. The full set of normal- and high-stress responses by horizon is reported in Table 7, Appendix A.6.2.

Robustness tests based on thresholds and subsamples confirm this caution. In some cases, the profiles retain the same sign across variants, but the correlations between profiles and the stability of the magnitude are not fully consistent. In particular, for cumulative returns, excluding the 2008–2009 period significantly alters some estimated profiles. For CISS, sign consistency decreases in certain specifications. Therefore, the regime section should be understood as an informative extension, not as the central result of the study.

7.6 Response regarding Growth vs. Value

The Growth-versus-Value component is the empirical result most directly linked to the central thesis of this study. While aggregate returns incorporate discount rate shocks, earnings revisions, sectoral composition, and monetary policy responses, the Growth-minus-Value differential allows us to examine

a dimension more closely aligned with the financial duration hypothesis. The dependent variable is defined as:

$$GV_t = r_t^{Growth} - r_t^{Value}$$

For cumulative periods, the following is used:

$$GV_{t,t+h} = \sum_{j=0}^{h} \left(r_{t+j}^{Growth} - r_{t+j}^{Value} \right)$$

The baseline specification estimates:

$$GV_{t,t+h} = \alpha_h + \beta_h \widehat{\varepsilon}_t^{\ell} + \Gamma_h' X_{t-1} + u_{t+h}$$

Under this convention, a coefficient $\beta_h > 0$ implies that a positive residual liquidity innovation is associated with higher cumulative returns for Growth relative to Value over the *h*-period horizon. This is the prediction derived from the financial duration channel: if Growth has a greater duration than Value, a compression in the discount rate should relatively favor Growth.

The results of the alternative balance-sheet-growth specification are consistent with this hypothesis. The estimated response peaks at $h = 4$, with a coefficient of $\beta = 0.0101$ and a *p*-value of 0.0022. The average response over horizons $h = 0$ to $h = 3$ is 0.0043; over horizons $h = 4$ to $h = 8$ it rises to 0.0084; and over horizons $h = 9$ to $h = 12$ it returns to 0.0043. In total, this specification presents 8 horizons significant at the 10% level and 5 horizons significant at the 5% level. The complete breakdown by horizon is provided in Appendix A.6.3 to keep the main body focused on the economic interpretation of the results.

The time profile is relevant. The response does not appear to be a purely contemporaneous effect, but rather a dynamic that strengthens over the short and medium term, particularly between four and seven months. This pattern is consistent with a gradual transmission channel from liquidity to discount rates, risk premiums, and relative portfolio reallocation. If the channel operated instantaneously and purely mechanically, one would expect a response concentrated at $h = 0$. Instead, the results suggest a slower dynamic propagation, consistent with portfolio frictions, expectation adjustment, and the gradual transmission of monetary conditions.

The robustness of the results supports this interpretation. The baseline residual-liquidity specification peaks at $h = 5$, with $\beta = 0.0092$ and $p = 0.006$. The specification with richer controls exhibits a peak of 0.0093 at $h = 5$. The specifications with two and three lags both peak at 0.0102 at $h = 5$. The post-2009 and post-2012 samples show higher peaks of 0.0110 and 0.0142, respectively, whereas excluding the COVID window yields a positive response with a peak of 0.0081. The alternative balance-sheet-growth specification peaks at 0.0101 at $h = 4$. The complete summary of specifications is presented in Appendix A.6.3.

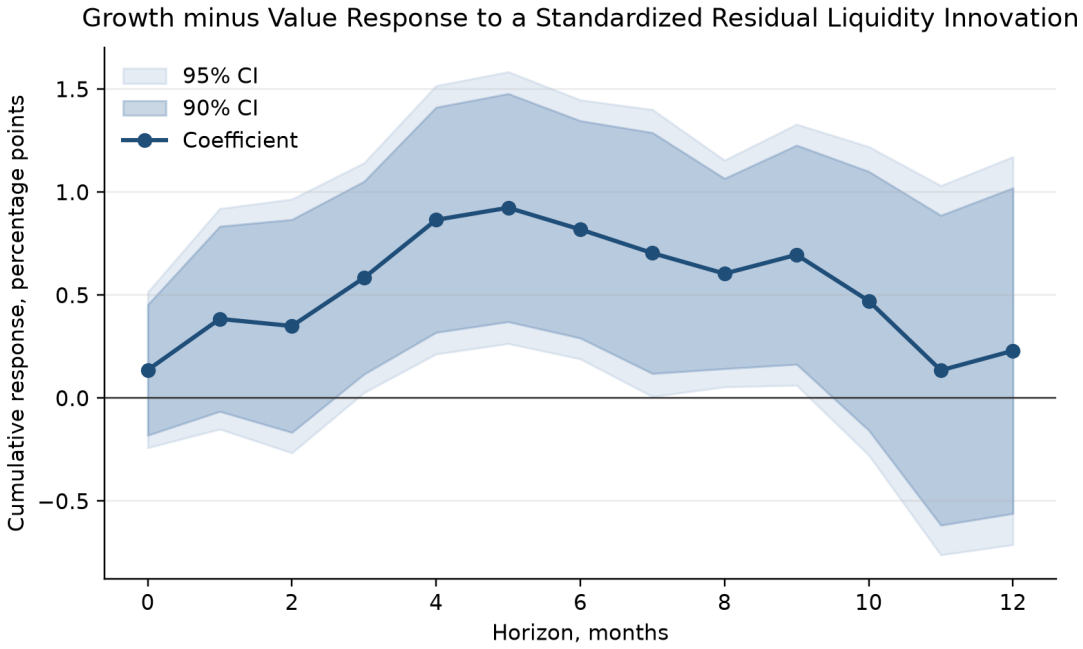


Figure 3

Cumulative response of Growth minus Value to a standardized residual liquidity innovation

Source: Author's own calculations and local-projection estimates based on ECB Data Portal data (European Central Bank, 2026b) and MSCI Europe Growth and MSCI Europe Value historical series obtained from Curvo (2026a, 2026b).

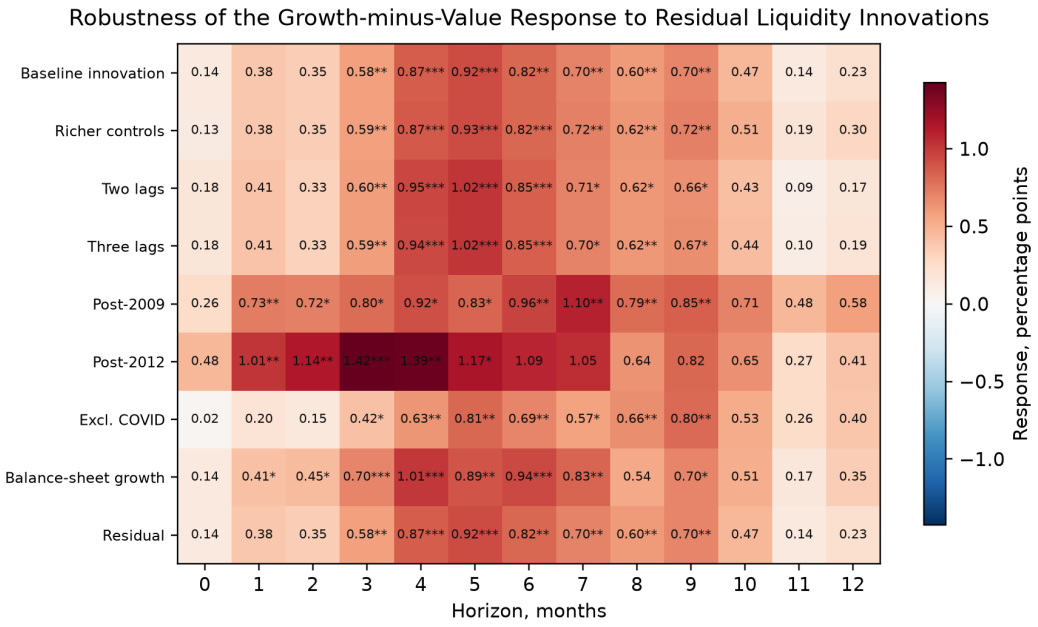


Figure 4

Robustness of the "Growth minus Value" response under alternative specifications

Source: Author's own calculations and robustness estimates based on ECB Data Portal data (European Central Bank, 2026b) and MSCI Europe Growth and MSCI Europe Value historical series obtained from Curvo (2026a, 2026b).

The economic interpretation is straightforward. If Growth assets derive a larger proportion of their value from expected cash flows at distant horizons, their implied financial duration will be longer than that of Value assets. In that case, a reduction in the relevant discount rate, whether from a compression of the risk-free component or a decline in risk premiums, should increase the present value of Growth by more. The estimated empirical result is consistent with this prediction: positive liquidity innovations are associated with subsequent relative outperformance of Growth relative to Value.

This interpretation should be viewed with caution. First, the analysis uses the MSCI Europe Growth and MSCI Europe Value indices, not the MSCI EMU Growth and MSCI EMU Value indices. Therefore, the findings should be interpreted as evidence from developed Europe consistent with the financial duration channel, not as a pure measure of the eurozone. Second, Growth versus Value does not exclusively capture financial duration. It may also reflect sector composition, currency differences, exposure to technology, healthcare, banks, energy, relative earnings shocks, or changes in global risk appetite. Third, the specification does not identify an exogenous liquidity shock, but rather a dynamic association conditional on the available controls.

The performance results within the Growth versus Value block do not support the conclusion that there is a robust amplification during periods of stress. The interaction coefficients do not consistently show precise horizons at the 10% level, and isolated signals under alternative thresholds do not generate a sufficiently consistent pattern. Therefore, the main conclusion of the block should focus on the baseline response and its robustness across liquidity measures, lags, and samples, rather than on a strong thesis of regime-driven amplification.

In summary, the Growth versus Value framework provides the evidence most consistent with the central hypothesis of this study. The positive and pointwise significant response at selected short- and medium-term horizons suggests that the Eurosystem's liquidity expansion is associated with a relative improvement in assets with longer financial durations compared to those with shorter financial durations. This evidence is consistent with the discount and risk premium compression channel, provided that a cautious and reduced-form interpretation is maintained.

7.7 Summary of Results

Taken together, the monthly sample (2000M1–2026M2) supports the design despite the unavailability of VSTOXX, APP/PEPP, and full-sample excess liquidity. The stationarity tests validate the transformation strategy, and the reaction function confirms that liquidity has a predictable component that cannot be treated as exogenous. Local projections of aggregate conditions are mixed: the CISS responds positively at some horizons and the cumulative EURO STOXX 50 response is negative, both consistent with intervention in stress rather than a causal effect of liquidity. Regime dependence is heterogeneous but not robust. The clearest result is the Growth-minus-Value response, positive and pointwise significant at short and medium horizons, peaking at $h = 5$ in the baseline residual-liquidity specification ($\beta = 0.0092$, $p = 0.006$) and at $h = 4$ in the alternative balance-sheet-growth specification ($\beta = 0.0101$, $p = 0.0022$), and qualitatively stable across robustness checks, consistent with the financial-duration channel under a cautious, reduced-form reading.

8. Discussion

8.1 Macrofinancial Analysis of the Results

The empirical results obtained in the previous chapter allow us to formulate a macrofinancial interpretation consistent with the central thesis of this study. The liquidity of the ECB and the Eurosystem does not appear as an exogenous variable that is mechanically and immediately passed on to asset prices, but rather as an endogenous variable linked to the evolution of financial conditions, systemic stress, and the institutional dynamics of European monetary policy itself. This characteristic is essential for correctly interpreting the results. In a monetary union such as the eurozone, the expansion of the Eurosystem's balance sheet cannot be understood solely as an aggregate monetary impulse, but also as a response to episodes of financial distress, market fragmentation, and the need to stabilize monetary transmission.

The evidence from the reaction function confirms this intuition. The evolution of liquidity exhibits a systematic component associated with its own persistence and lagged financial variables. Therefore, part of the expansion in liquidity should be interpreted as the Eurosystem's response to adverse conditions, rather than as an autonomous monetary shock. This result justifies the strategy followed in this study: separating, to the extent possible, the predictable component of liquidity from the residual innovation subsequently used in local projections. Residual liquidity innovation does not constitute an exogenous monetary shock in the strict sense, but it helps reduce confusion between the policy response and subsequent transmission.

From this perspective, the first interpretive conclusion is that the relationship between liquidity and asset prices must be viewed dynamically. A negative contemporary correlation between liquidity and financial returns does not necessarily imply that liquidity is causing market conditions to deteriorate. Rather, it may reflect the fact that the central bank expands its balance sheet when the financial environment has already deteriorated. This distinction is central to the analysis of unconventional monetary policy, because central bank interventions typically occur precisely at times when risk premiums rise, stock market valuations fall, or market liquidity deteriorates. Consequently, interpreting the estimated coefficients requires distinguishing between the timing of the intervention and its subsequent dynamic effects.

The second conclusion is that liquidity transmission does not appear to operate uniformly across all financial variables. The results regarding aggregate financial conditions are less clear than those in the Growth versus Value section. The responses of the CISS and the aggregate returns of the EURO STOXX 50 suggest that a significant portion of the dynamics is driven by the Eurosystem's stabilizing role. In contrast, the response of the Growth minus Value spread is more consistent with the central hypothesis of this study: liquidity expansions are associated, over certain time horizons, with a relative improvement in assets with longer financial durations relative to those with shorter financial durations.

This distinction between aggregate and relative outcomes is significant. The hypothesis of this study does not simply argue that liquidity uniformly raises all asset prices. The central proposition is more specific: if liquidity compresses discount rates and risk premiums, then it should favor those assets whose valuations depend more heavily on expected cash flows over long-term horizons. Therefore, the most important result is not necessarily the response of the aggregate stock market index, but rather the relative response between Growth and Value. This response allows us to examine whether the liquidity architecture affects the intertemporal composition of financial valuation, not just the aggregate

level of prices.

A comprehensive reading of Chapter 7 is therefore as follows. First, Eurosystem liquidity is endogenous and responds to financial conditions; second, the aggregate transmission to market stress, volatility, or returns is conditioned by this endogeneity and yields mixed result; third, the Growth versus Value section provides a clearer signal that is more aligned with the theoretical mechanism of financial duration; fourth, this evidence should be interpreted cautiously, as a dynamic association consistent with the discount channel, not as a structural demonstration of direct causality.

The interpretive contribution of this work lies precisely in this combination. Monetary liquidity appears as a mechanism for short-term financial stabilization, but also as a force capable of altering relative prices over longer horizons. This dual nature allows us to link unconventional monetary policy with asset pricing theory. The same intervention that reduces financial tensions can also alter how markets discount future cash flows. Hence, the discussion must be situated at the intersection of monetary economics, macrofinance, and asset pricing.

8.2 Liquidity, Financial Stability, and Asset Prices

The first aspect of these results is the stabilizing role of liquidity. Within the institutional framework of the euro area, the ECB and the Eurosystem act not only as authorities that set short-term interest rates, but also, as providers of liquidity in a potentially fragmented financial system. Balance sheet expansion, refinancing operations, asset purchase programs, and ample reserves are all part of an architecture designed to preserve monetary transmission, reduce funding strains, and prevent disorderly deleveraging dynamics.

This stabilizing role is particularly important because the euro area combines a common monetary policy with heterogeneous sovereign markets, banking systems and national financial conditions. During periods of stress, the same monetary policy stance may be transmitted unevenly if sovereign spreads widen, market liquidity deteriorates or certain jurisdictions face more severe financing constraints. In this context, liquidity provision affects not only the aggregate level of financial conditions but also the functioning of the transmission mechanism.

The empirical results are consistent with this interpretation. Eurosystem liquidity exhibits a component that reacts to financial conditions, suggesting that balance sheet expansions tend to occur primarily in times of stress. This explains why the contemporaneous responses of some financial variables may be ambiguous. If the central bank intervenes when stress is already high, the expansion of liquidity may coincide with increases in the CISS, stock market declines or a deterioration in aggregate returns. This temporal coincidence should not be confused with a negative causal effect of liquidity; rather, it reflects the fact that unconventional monetary policy often responds to prior market deterioration.

From an asset valuation perspective, financial stabilization affects the price through the risk premium. If the relevant discount rate is defined as

$$R_t = r_t^f + \pi_t \tag{18}$$

where (r_t^f) represents the risk-free component and (π_t) the risk premium, a stabilizing intervention can reduce (R_t) even if it does not immediately alter expected cash flows. The reduction in (π_t) can occur through several channels: lower systemic stress, improved market liquidity, reduced fragmentation risk, spread compression and greater investor willingness to hold risky assets. This mechanism con-

nects directly to the literature on asset purchases, portfolio rebalancing, and risk premium compression Altavilla et al. (2015) and Krishnamurthy and Vissing-Jorgensen (2011).

However, the aggregate results of this study suggest that this stabilization should not be interpreted as a simple linear relationship. The response of the cumulative returns of the EURO STOXX 50 to residual liquidity shocks does not provide an immediate and uniform positive signal. This result does not invalidate the study's hypothesis. In contrast, it reinforces the need to distinguish between the aggregate effect on the stock market and the relative effect on assets with different financial durations. The aggregate index combines companies, sectors, cyclical exposures, interest rate sensitivities and risk premiums that are highly heterogeneous; therefore, an expansion of liquidity may have opposing effects within the index, such that the aggregate result masks significant differences between investment styles.

Financial stabilization may also have a regime-dependent dimension. In normal times, additional liquidity expansion may have limited marginal effects if risk premiums are already low and markets are functioning normally. In contrast, during periods of stress, the same expansion can have more intense effects because it acts on balance sheet constraints, liquidity strains, risk aversion and refinancing risk. This intuition justifies the regime-interaction specifications used in the previous chapter. The goal is not to assert that liquidity always has the same effect, but rather to examine whether its transmission varies across initial financial states.

The economic interpretation of this section is, therefore, cautious. Eurosystem liquidity plays a stabilizing role, but this role is observed in an environment of strong endogeneity. The monetary authority acts when the system is under stress, and for this reason, the estimated coefficients should not be interpreted as laboratory responses to purely exogenous shocks. The evidence suggests that liquidity is part of a macrofinancial containment mechanism, but it does not allow for the complete isolation of all the channels through which that containment is transmitted to aggregate asset prices.

This caution does not diminish the analysis's relevance; rather, it provides a better understanding of the euro area's liquidity architecture. Unconventional monetary policy does not operate solely through the level of official interest rates. It also operates through balance-sheet provision, the reduction of financial dislocations, the central bank's absorption of assets, and the modification of the premiums demanded by investors. These channels are precisely what set the stage for the central question of this study: whether financial stabilization can simultaneously alter the relative valuation of assets with different financial durations.

8.3 Financial Duration and Relative Revaluation

The most significant finding of this study is found in the Growth versus Value section. Unlike the aggregate responses, the response of the Growth minus Value spread shows a pattern more consistent with the financial duration hypothesis. In the baseline residual-liquidity specification, the cumulative response of the spread is positive across several horizons, peaks at short-to-medium horizons (around the fifth month), and is pointwise statistically significant at selected horizons between the third and ninth month. The robustness specifications preserve the same general direction of the result, reinforcing the interpretation that liquidity is associated with a relative improvement of Growth relative to Value.

The economic interpretation of this result must be based on the asset pricing equation. For a portfolio (i), the price can be expressed as

$$P_t^i = \mathbb{E}_t \left[\sum_{j=1}^{\infty} \frac{CF_{t+j}^i}{(1 + R_t^i)^j} \right]$$

where (CF_{t+j}^i) represents the expected cash flows and (R_t^i) the relevant discount rate. The percentage sensitivity of the price to the discount rate can be approximated using the financial duration:

$$\Delta \ln P_t^i \approx -D_t^i \Delta R_t^i$$

If Growth assets have a longer financial duration than Value assets, then

$$D_t^G > D_t^V$$

Given a common or highly correlated compression of the discount rate, the expected difference in responses is

$$\Delta \ln P_t^G - \Delta \ln P_t^V \approx -(D_t^G - D_t^V)\Delta R_t$$

If $(\Delta R_t < 0)$, the above expression will be positive. In other words, a reduction in the discount rate should give Growth a relative advantage over Value. This is the central prediction that the empirical section seeks to test.

The results are consistent with this prediction. The positive response of $(GV_{t,t+h})$ suggests that, following liquidity expansions or residual liquidity innovations, growth assets tend to outperform value assets over the short and medium term. This dynamic is consistent with a discount channel: as risk-free rates, risk premiums, or both narrow, the present value of future cash flows increases; and this increase is greater for assets whose valuation depends more heavily on distant expected earnings.

This interpretation does not require us to claim that Growth is a perfect measure of financial duration. Growth and Value portfolios differ for many reasons: sector composition; exposure to technology, banking, healthcare, or energy; differences in expected returns; cyclical sensitivity; earnings expectations; dividends; liquidity; and geographic or currency factors. However, from an asset-pricing perspective, Growth serves as a reasonable proxy for assets with a greater relative weight to future cash flows, while Value approximates assets more closely linked to present earnings, existing assets, or book values. The hypothesis does not require that every movement in the Growth minus Value spread be explained by duration, but rather that the duration component be sufficiently relevant to appear in the dynamic response to liquidity.

This clarification is important because it prevents an overly mechanistic interpretation. The (GV_t) spread is not a pure measure of duration. It may also capture sectoral shocks, changes in earnings expectations, or market rotations not directly related to the ECB; therefore, the correct conclusion is not that liquidity fully explains the relative performance of Growth versus Value; but rather that the observed evidence is consistent with the financial duration channel. Empirically, a positive and significant response of the spread to liquidity innovations supports the hypothesis that assets with longer durations are more sensitive to discount compression.

The robustness of the result is particularly noteworthy. Alternative specifications continue to show, although with differences in magnitude, a positive response from Growth minus Value. This suggests that the result does not depend exclusively on a single set of parameters. The persistence of the signal over medium-term horizons is consistent with the idea that liquidity transmission to relative prices need

not be instantaneous. Investors may adjust portfolios gradually, risk premiums may compress with a lag and the valuation of future cash flows may react as expectations regarding interest rates, liquidity and financial stability are updated.

The Growth versus Value result also helps explain why aggregate responses may be weaker or more ambiguous. A broad stock market index combines assets with different financial durations. If liquidity favors some assets more than others, the aggregate effect may be partially offset. The cross-sectional comparison reveals a dimension that the aggregate index does not capture: not only how much the market rises or falls, but which types of assets appreciate relatively more. This is one of the main contributions of this study.

The macrofinancial implication is clear. Eurosystem liquidity can affect the valuation structure of markets, not just at the aggregate level. When the discount rate narrows, the market does not revalue all future cash flows uniformly. It revalues those cash flows that are further in the future more significantly. Therefore, an expansionary liquidity policy may favor long-duration financial assets even when its effect on the aggregate index is ambiguous. This distinction between aggregate and relative effects constitutes the interpretive core of this chapter.

8.4 Intertemporal Fragility and the Accumulation of Vulnerabilities

The evidence presented above allows us to introduce a broader discussion of intertemporal fragility. The concept of intertemporal fragility does not imply that every expansion of liquidity necessarily leads to a bubble; nor that the relative appreciation of growth stocks is in itself a misallocation of capital. This discussion is framed as a theoretical interpretation compatible with the results, not as an empirical finding of this thesis. Its meaning is more precise: if liquidity persistently compresses discount rates and risk premiums, it can increase the relative valuation of assets whose justification depends on distant cash flows, a mechanism that connects to the literature on the endogenous build-up of financial vulnerabilities during prolonged periods of accommodative conditions (Borio, 2014). In that case, the valuations of long-duration assets may become more sensitive to a subsequent reversal in discount rates or risk premia; whether this translates into systemic vulnerability depends on conditions such as leverage, balance-sheet constraints or interconnections that this thesis does not estimate.

This idea helps link financial stabilization to the accumulation of vulnerabilities. In the short term, liquidity provision can stabilize markets. It can reduce refinancing pressures, contain episodes of stress, improve market functioning, and prevent disorderly fire sales. From this perspective, liquidity fulfills a necessary macrofinancial function, especially in a monetary union at risk of fragmentation. However, the same intervention can affect relative prices. By reducing the relevant discount, it favors assets with longer financial durations to a greater extent; thus, immediate stabilization can coexist with greater future sensitivity to changes in monetary conditions.

The mechanism follows directly from the present-value framework introduced earlier in this study. Holding expected cash flows fixed, a decrease in R_t raises the present value of the asset, and the further out those cash flows lie, the greater the price sensitivity to changes in R_t. This ceteris paribus condition is important; the duration channel operates through the discount rate, and its sign can be reversed if a monetary expansion simultaneously triggers an adverse revision of expected future cash flows for long-duration assets relative to short-duration ones, a possibility examined in Chapter 3. Consequently, when market valuations become increasingly conditioned on a low-discount-rate environment, the valuations of assets with longer durations may experience a particularly sharp rise. This

rise is not necessarily irrational; it may be fully consistent with the valuation model under a low discount rate. Fragility arises only if that valuation depends on the persistence of exceptionally favorable monetary and financial conditions.

From this perspective, the Growth versus Value result can be interpreted as evidence of a possible relative reallocation of valuation toward assets that are more dependent on the future. The study does not demonstrate that such a reallocation is inefficient, but it does show that liquidity may be associated with changes in the implied time structure of financial prices. This distinction is fundamental. For the limited interpretation adopted here, intertemporal fragility does not require evidence of actual capital misallocation. It refers instead to the possibility that relative valuations become increasingly dependent on the persistence of low discount rates.

The literature identifies several channels through which such vulnerabilities may, in principle, accumulate; this thesis does not directly estimate any of them. A first channel operates through risk-taking: a persistent narrowing of risk premiums may encourage greater risk-taking and exposure to higher-yielding assets (Borio & Zhu, 2012; Jiménez et al., 2014). It should be noted that this evidence pertains to bank lending and credit risk-taking, not to equity duration; it is cited here as a theoretical motivation for the mechanism, not as direct evidence on the Growth minus Value spread. A related dimension concerns the procyclicality of leverage and balance-sheet capacity documented specifically among market-based intermediaries and broker-dealers, whose marked-to-market balance sheets expand when asset prices rise and contract sharply in downturns (Adrian & Shin, 2010); again, this result applies to intermediary balance sheets and cannot be extrapolated directly to equity relative valuation. A second asset-pricing channel is that the relative appreciation of long-duration assets increases their sensitivity to a future reversal of discount rates or risk premiums, Lettau and Wachter (2007) provides a duration-based framework for the differential exposure of growth and value assets to discount-rate shocks; while Gormsen and Lazarus (2026) offers more direct evidence on the sensitivity of long-duration equities to pure discounting shocks. Finally, if relative valuation feeds back into corporate financing conditions, it might indirectly affect the sectoral allocation of capital, but this thesis does not estimate that real channel.

This interpretation helps make sense of the apparent tension between stabilization and vulnerability. Liquidity can have a stabilizing effect at the time of intervention and, at the same time, contribute to a price structure that is more dependent on the continuation of a low discount. There is no contradiction between these two statements. The first refers to the short-term effects on stress, market liquidity, and refinancing; the second refers to the medium-term effects on relative valuation, sensitivity to the discount rate, and exposure to monetary reversals. Chapter 7 suggests precisely that the clearest effect appears in the relative dimension of Growth versus Value, not necessarily in the market's immediate aggregate response.

The concept of intertemporal fragility also helps avoid an overly prescriptive interpretation. It does not assert that expansionary monetary policy is necessarily harmful, nor that financial stabilization is undesirable. During periods of stress, a lack of liquidity can lead to disorderly contractions, a loss of financial intermediation and macroeconomic deterioration. The point is more limited and more analytical, stabilization through discount compression can alter the relative signals by which the market values present and future cash flows. Hence, a comprehensive assessment of liquidity policy must consider not only its ability to reduce stress, but its effects on the time structure of asset valuation.

This is one of the conceptual contributions of this study. Eurosystem liquidity is not analyzed solely as a monetary aggregate or as an instrument of aggregate stabilization. It is analyzed as a variable

that can affect the price of time in financial markets. If the discount rate determines the relative value of near- and long-term cash flows, then a policy that persistently alters that rate can alter the intertemporal composition of valuations. In this sense, the Growth versus Value channel allows for the empirical observation of a dimension that is usually hidden in aggregate analyses of monetary policy.

8.5 Causal Scope of the Results

Despite the consistency between the results and the theoretical hypothesis, the causal implications of this study should be interpreted with caution. The empirical strategy does not identify a pure exogenous monetary shock in a structural, narrative, or high-frequency sense. The residual liquidity innovation is derived from a reaction function estimated from observed data, but it does not completely eliminate endogeneity. Therefore, the estimated responses should be interpreted as conditional dynamic associations, not as definitive causal effects of liquidity on relative prices.

The main source of caution is the endogeneity of monetary policy. The ECB and the Eurosystem inject liquidity in response to economic and financial conditions. Although the reaction function accounts for part of this dynamic, there may always be relevant information that the researcher fails to observe but that the central bank or markets take into account. For example, expectations of future deterioration, emerging banking tensions, fragmentation risks, or market liquidity signals may simultaneously influence the Eurosystem's actions and asset valuations. In that case, the residual of the reaction function would continue to incorporate components that are not fully exogenous.

The second source of caution is the simultaneity among liquidity, interest rates, risk premiums, and asset prices. Liquidity can affect risk premiums, but risk premiums can also trigger changes in liquidity. Asset prices may respond to monetary policy expectations before the balance sheet expansion materializes. Furthermore, the Growth versus Value response may reflect changes in relative earnings expectations, sector composition, or global factors not fully captured by the controls. Therefore, even a positive response in the Growth minus Value spread should not be interpreted as definitive proof of direct causality.

The third source of caution relates to the reduced-form nature of the local projections. Local Projections estimate horizon-specific responses of a dependent variable to a shock, conditional on controls. This methodology is flexible and suitable for studying time dynamics Jordà (2005), but it does not, in and of itself, replace a structural identification strategy. In particular, the estimated responses do not perfectly distinguish between the risk-free rate channel, changes in risk premia, market-liquidity effects, portfolio rebalancing, signaling about the expected future path of monetary policy, and revisions to expected future cash flows. These mechanisms may operate simultaneously, and the Local Projections neither identify their separate contributions nor establish that the estimated Growth minus Value response is driven exclusively by discount-rate news.

For these reasons, the interpretation must remain deliberately cautious. The results do not demonstrate that the ECB directly causes a re-rating of Growth relative to Value. Nor do they prove that monetary policy has generated a bubble in long-duration assets. The defensible conclusion is different: the observed dynamic patterns are consistent with a financial duration channel in which liquidity expansions are associated with a relative improvement in assets that are more sensitive to discounting.

This caution does not weaken the work; rather, it strengthens it. The contribution of the analysis lies in empirically organizing a complex relationship and showing that the results are consistent with a well-defined theoretical hypothesis. The evidence is particularly relevant because the Growth versus

Value result emerges from a dynamic specification, with controls and robustness against reasonable alternatives. However, the inference must be situated in the realm of empirical compatibility, not strong causal demonstration.

A possible future extension would involve strengthening the identification process by incorporating high-frequency monetary policy events, surprises in ECB announcements, or external instruments based on unanticipated changes in asset purchases. More direct measures of expected interest rates, term premiums, European corporate spreads, or the sectoral composition of Growth and Value portfolios could also be incorporated. These extensions would allow for a more precise separation of the various transmission channels. However, in this paper, the objective is more limited: to analyze whether the observed dynamic responses are consistent with the theoretical framework of liquidity, discount, and financial duration.

8.6 Empirical and Methodological Limitations

The analysis has several limitations that should be considered when interpreting the results. The first and most important is the geographical limitation of the Growth versus Value proxy. In the absence of operational, consistent, and sufficiently long time series for the MSCI EMU Growth and MSCI EMU Value indices, this study uses the MSCI Europe Growth and MSCI Europe Value indices. This decision allows us to capture the cross-sectional dimension of Growth versus Value, but it means that the empirical sample does not strictly represent the eurozone. MSCI Europe includes developed European markets that may not be part of the euro area, so the results should be interpreted as European evidence consistent with the financial duration channel, not as a pure measure of relative valuation within the monetary union.

The second limitation is the sectoral composition of Growth and Value portfolios. The differences between the two styles are not limited solely to financial duration. Growth portfolios may have a higher weighting in sectors such as technology, healthcare, or consumer discretionary, while Value portfolios may be more exposed to banking, energy, industrials, or sectors with high tangible asset intensity. Therefore, part of the explanation for the spread may reflect sectoral shocks or changes in relative earnings expectations. The interpretation as financial duration is theoretically defensible, but not exclusive. A natural extension would be to control for European sector indices or to decompose the Growth minus Value spread by sector.

The third limitation concerns the measurement of liquidity. The Eurosystem's baseline total assets variable captures a broad measure of balance sheet expansion, but does not perfectly distinguish between refinancing operations, asset purchase programs, valuation changes, maturities, reinvestments, or accounting adjustments. More specific variables such as excess liquidity, the APP, or the PEPP could capture more specific channels, but their availability, time consistency, or sample coverage may be inferior. Therefore, the study uses a broad and reproducible balance-sheet measure, supplemented with robustness analysis where possible. More granular liquidity measures are either unavailable or insufficiently homogeneous over the full sample: operationally consistent APP and PEPP series cannot provide a continuous program-specific measure from 2000 to 2026, while the available excess-liquidity series covers a substantially shorter period. In addition, the absence of a continuous VSTOXX series prevents the analysis from incorporating an equity-market implied-volatility control alongside the broader CISS indicator. These constraints limit the decomposition of liquidity interventions and financial stress, although they do not prevent the estimation of the baseline

specification.

The fourth limitation is the difficulty of separating transmission channels. Liquidity can affect asset prices through risk-free rates, term premiums, credit risk premiums, market liquidity, monetary policy expectations, signaling, and portfolio rebalancing. The estimated specifications allow for the assessment of dynamic responses, but they do not separately identify each of these channels. Consequently, when interpreting a positive response of Growth minus Value as consistent with discount compression, it must be understood that such a discount may incorporate both the risk-free component and various risk premiums.

The fifth limitation is the potential for regime instability. The 2000M1–2026M2 sample covers a wide range of distinct episodes: the period leading up to the global financial crisis, the sovereign debt crisis, the era of negative interest rates, the pandemic, the PEPP, the surge in inflation, and the subsequent monetary normalization. The relationship between liquidity, financial stress, and asset prices may not be constant throughout the entire period. Regime interactions partially capture this heterogeneity, but do not completely rule out the possibility of structural changes in monetary transmission. Consistent with this, the regime-dependence results (H3) are imprecisely estimated and not robust across specifications, so the regime interactions should be regarded as exploratory rather than confirmatory.

The sixth limitation relates to the COVID-19 episode. The pandemic triggered extreme fluctuations in liquidity, financial stress, asset prices, and central bank policy. These months are informative because they represent a pivotal episode in the European liquidity architecture, but they may also disproportionately influence the estimates. Specifications that exclude the COVID window help assess this issue, but they do not replace a comprehensive structural analysis of regime shifts.

The seventh limitation is the effective sample size. Although the monthly dataset provides sufficient breadth for the analysis, the number of observations remains limited for estimating dynamic models with many controls, interactions, and time horizons. This constraint suggests avoiding overly parameterized specifications.

A further inferential consideration concerns the use of an estimated residual and the assessment of responses across horizons. The residual liquidity innovation is obtained from an estimated first-stage reaction function. For residual-generated regressors, conventional second-stage inference can remain asymptotically valid under appropriate regularity conditions, but the present analysis does not formally establish those conditions or propagate first-stage estimation uncertainty through a joint resampling procedure (Pagan, 1984). This issue should therefore be treated as a finite-sample caution rather than as evidence that the reported HAC standard errors are necessarily downward biased. In addition, the confidence intervals are pointwise by horizon. They support statements about individual coefficients at specific horizons, but not a joint confidence statement for the complete response path; simultaneous confidence bands or joint tests would be required for the latter (Montiel Olea & Plagborg-Møller, 2019). Accordingly, the Growth minus Value response should be described as pointwise significant at selected horizons, rather than jointly significant across the full response profile.

Finally, there is a general limitation to the interpretation. This study examines financial valuation, not actual direct investment. Although the theoretical framework allows for a connection between financial duration, discounting, and the relative allocation of capital, the empirical results do not prove that liquidity has led to actual changes in the eurozone's productive structure. To confirm this channel, it would be necessary to analyze business investment, corporate financing, the sectoral composition of capital, productivity, or microeconomic data from firms. This study is limited to examining the relative

valuation of financial assets and its compatibility with a duration hypothesis.

Taken together, these limitations do not invalidate the analysis, but they do limit its scope. The contribution of this study is to show that the Eurosystem's liquidity architecture can be examined not only in terms of its aggregate effects on stress or stock market returns, but also in terms of its potential impact on the relative prices of assets with different financial durations. The results support a cautious interpretation: European liquidity appears to act as a financial stabilization mechanism and, at the same time, exhibits a dynamic association with the relative appreciation of Growth assets relative to Value assets. This dual dimension forms the core of the discussion and sets the stage for the final answer to the research question in the conclusions chapter 9.

9. Conclusions

9.1 Answer to the Research Question

This undergraduate thesis has analyzed whether the ECB's and the Eurosystem's liquidity expansions favor the relative appreciation of assets with longer financial durations, approximated by Growth portfolios; compared to assets with shorter financial durations, approximated by Value portfolios. The research question lies at the intersection of European monetary policy, financial conditions and the intertemporal valuation of assets. The central question was not merely whether Eurosystem liquidity raises the aggregate price of financial assets, but whether it can also alter the relative valuation structure among assets with differing discount rate sensitivities.

The answer derived from the empirical analysis is affirmative, though necessarily qualified. The results provide evidence that positive residual liquidity innovations are associated with the relative outperformance of Growth over Value at selected short and medium-term horizons. In particular, the Growth minus Value response is positive and pointwise statistically significant at selected horizons, reaching its baseline peak at $h = 5$ ($\beta = 0.0092$, $p = 0.006$), while the alternative balance-sheet-growth specification peaks at $h = 4$ ($\beta = 0.0101$, $p = 0.0022$), and remains qualitatively positive across the main robustness specifications. This pattern is consistent with the financial-duration channel, provided that the relevant innovation operates partly through discount-rate compression and is not offset by adverse revisions to the expected cash flows of Growth relative to Value.

The empirical findings are interpreted through the financial-duration relation developed in Chapter 3. Under that framework, a compression of the relevant discount rate increases the present value of longer-dated cash flows more strongly, conditional on the absence of offsetting adverse cash-flow news. The estimated sign and timing of the Growth minus Value response are consistent with this mechanism, although the empirical design does not structurally identify it or exclude alternative explanations based on relative cash-flow expectations, sector composition, or global shocks.

This conclusion should not be interpreted as an assertion of strong structural causality. The empirical design used mitigates part of the endogeneity problem through a Eurosystem reaction function and the use of residual liquidity innovations, but it does not identify a fully exogenous monetary shock in a narrative, institutional, or high-frequency sense. Therefore, the correct conclusion is not that the ECB directly causes a rise in Growth, nor that it has generated a bubble in assets with longer financial durations. The appropriate conclusion is that, after conditioning on the estimated reaction function, which only partially mitigates the underlying endogeneity, positive residual liquidity innovations are associated with a relative repricing in favor of the higher-duration proxy. This association is consistent with the financial-duration channel, but it does not constitute structural identification of the discount-rate mechanism.

It should also be noted that the answer to the research question depends on the proxy used. The Growth versus Value analysis is based on the MSCI Europe Growth and MSCI Europe Value indices, not on the MSCI EMU Growth and MSCI EMU Value indices. This decision was made for reasons of data availability and allows us to capture the cross-sectional dimension of financial duration, but it limits the strictly eurozone interpretation of the results. Consequently, the evidence should be understood as European evidence consistent with the duration channel, not as a pure measure of relative valuation within the euro area.

9.2 Evaluation of the Hypotheses

The first hypothesis posited that Eurosystem liquidity responds endogenously to financial conditions. This hypothesis is supported by the reaction function results. The evolution of liquidity does not appear to be a purely exogenous process, but rather a persistent variable that is partially explained by lagged financial conditions. This result is fundamental to the interpretation of the study as a whole. If central bank liquidity increases precisely when financial conditions deteriorate, a contemporaneous correlation between liquidity expansion and market stress cannot be interpreted as evidence that liquidity causes financial deterioration. Rather, it may reflect the Eurosystem's stabilizing response to a stressed environment.

From a methodological standpoint, this conclusion validates the empirical framework adopted. The reaction function allows us to distinguish between the systematic component of monetary policy and a residual liquidity innovation:

$$\Delta \ell_t = \alpha + \sum_{i=1}^{p} \rho_i \Delta \ell_{t-i} + \sum_{i=0}^{q} \gamma_i' S_{t-i} + \varepsilon_t^{\ell}$$

The term (ε_t^{ℓ}) does not represent a perfect exogenous shock, but rather a change in liquidity that is less contaminated by the central bank's predictable response to observable conditions. Therefore, the first hypothesis not only receives empirical support but also becomes a central component of the study's identification design.

The second hypothesis posited that liquidity innovations are associated with dynamic improvements in financial conditions. Local projections of the CISS, interest rates, the yield curve slope and aggregate stock returns do not show a clear pattern of immediate improvement. Over some horizons, the CISS responds positively and the cumulative returns of the EURO STOXX 50 exhibit a negative conditional association with residual liquidity innovation. These results, far from necessarily invalidating the theoretical framework, reinforce the importance of endogeneity: even after estimating a reaction function, residual innovation may still be associated with episodes of financial deterioration not fully captured by the included controls.

Therefore, H2 is not supported in its original directional formulation. The estimates reveal statistically relevant dynamic associations between residual liquidity innovations and aggregate financial conditions, but they do not show a uniform subsequent improvement: CISS rises at selected horizons, the EURO STOXX 50 response is negative and the responses of interest rates and the yield curve are comparatively weak. The evidence should therefore be classified as mixed and directionally inconclusive rather than as indirect confirmation of a stabilizing effect. This relationship is deeply influenced by the state of the financial system, the central bank's institutional response, and the simultaneity of market stress and monetary intervention.

The third hypothesis posited that liquidity transmission depends on the financial regime, being more intense during periods of high stress. The results provide some evidence of heterogeneity, but not a robust and general confirmation. Some estimated responses differ between normal and stress regimes, but the signs, magnitudes and time profiles are not sufficiently stable to make this dimension the main conclusion of the study. Consequently, the regime-dependence hypothesis is not robustly supported: the evidence is exploratory and inconclusive rather than confirmatory. This absence of robust confirmation does not amount to a rejection of the underlying theoretical possibility of nonlinear transmission.

This interpretation is methodologically sound. The regime dependence of unconventional monetary policy is difficult to estimate precisely, especially in relatively limited monthly samples and given the highly heterogeneous nature of stress episodes. The global financial crisis, the sovereign debt crisis, the period of negative interest rates, the pandemic and the subsequent phase of monetary normalization do not represent the same type of financial regime. Therefore, although the idea of nonlinearity is theoretically plausible, the evidence obtained suggests treating it as an interpretive extension rather than the study's central finding.

The fourth hypothesis constituted the study's core distinction. This hypothesis posited that liquidity expansions favor assets with longer financial durations, approximated by Growth, over those with shorter financial durations, approximated by Value. H4 receives the strongest empirical support among the four hypotheses. The estimated Growth minus Value responses are positive and pointwise statistically significant at selected short and medium-term horizons, while their qualitative sign remains stable across the main robustness specifications. In the baseline residual-liquidity specification, the response reaches its maximum at $h = 5$ ($\beta = 0.0092$, $p = 0.006$), where the estimate is positive and pointwise statistically significant, while the alternative balance-sheet-growth specification peaks at $h = 4$ ($\beta = 0.0101$, $p = 0.0022$). This finding should nevertheless be interpreted with several qualifications: no joint test establishes significance of the response path as a whole; MSCI Europe Growth and MSCI Europe Value cover developed European markets rather than the euro area exclusively; and the Growth minus Value spread is an imperfect financial-duration proxy that does not isolate discount-rate news from cash-flow news, sector composition or other common shocks.

In terms of local projections, the result can be expressed as:

$$GV_{t,t+h} = \alpha_h + \beta_h \varepsilon_t^{\ell} + \Gamma_h' X_{t-1} + u_{t+h}$$

where the central hypothesis requires $(\beta_h > 0)$ over the relevant time horizons. The evidence obtained is consistent with this condition. Consequently, the fourth hypothesis is cautiously supported: not as definitive structural proof, but as dynamic evidence consistent with the financial duration interpretation.

The overall assessment of the hypotheses is therefore mixed. The hypothesis of liquidity endogeneity is supported within the estimated reaction-function specification. H2 yields mixed and directionally inconclusive evidence and does not confirm a uniform improvement in aggregate financial conditions. The hypothesis of regime dependence is not robustly supported and should be regarded as exploratory. The hypothesis of relative outperformance of Growth over Value receives the strongest empirical support and constitutes the main contribution of this study.

9.3 Implications for European Monetary Policy and Asset Pricing

The first implication of this study is that European monetary policy should be assessed not only by its aggregate effects on liquidity, financing conditions, and broad market indices, but also by its effects on relative asset valuations. The empirical results suggest that the transmission of Eurosystem liquidity may differ according to the temporal distribution of expected cash flows. Consequently, an assessment based solely on the stock market's aggregate response may overlook economically relevant changes in the internal composition of valuations. From this perspective, monetary policy influences not only the overall price level of financial assets, but potentially also the relative price assigned to near-term and distant expected cash flows.

This relative dimension is particularly important in the context of unconventional monetary policy. Balance-sheet expansion, asset purchases, and abundant central bank liquidity may operate through the compression of risk-free rates, risk premia, liquidity premia, and portfolio rebalancing incentives. Their effects therefore need not be distributed uniformly across assets. The positive response of the Growth minus Value differential at selected short and medium-term horizons is consistent with the possibility that assets with greater exposure to distant expected cash flows respond more strongly to changes in the relevant discount environment. The policy implication is not that aggregate stabilization is unimportant, but that it provides only one dimension of monetary transmission.

At the same time, interpreting these effects requires explicit treatment of policy endogeneity. As discussed in subsections 8.1, 8.2 and 8.5, Eurosystem liquidity frequently expands in response to deteriorating financial conditions, market fragmentation, or threats to monetary transmission. The distinction between this systematic stabilizing response and subsequent liquidity transmission must therefore remain central to any empirical assessment. The aggregate and relative effects identified in this study should be interpreted within that endogenous institutional setting rather than as the consequences of a purely exogenous monetary intervention.

The institutional configuration of the euro area reinforces this conclusion. Unlike a fully integrated monetary and fiscal union, the eurozone combines a common monetary policy with heterogeneous sovereign markets, national banking systems and jurisdiction-specific financing conditions. Eurosystem liquidity consequently performs both an aggregate monetary function and a transmission-preserving function. Policies such as asset purchases or flexible reinvestments can influence financial conditions while simultaneously attempting to contain fragmentation. This institutional specificity means that the effects of liquidity cannot be understood solely through a representative euro-area asset or a single aggregate market index. They must also be examined across countries, market segments, sectors and investment styles.

For asset-pricing analysis, financial duration provides a useful framework for organizing this heterogeneity. As developed in subsection 8.3, the sensitivity of an asset to changes in discount rates depends partly on the timing of its expected cash flows. The Growth versus Value comparison used in this study operationalizes that insight imperfectly but informatively: it allows monetary transmission to be examined through relative discount sensitivity rather than only through aggregate returns. This perspective connects monetary policy to the intertemporal structure of valuation by asking how changes in financial conditions alter the relative weights assigned to present and future cash flows.

A more granular research and policy framework should therefore examine monetary transmission across investment styles, sectors and different implied maturities of cash flows. Relevant dimensions include sectoral composition, reliance on intangible capital, earnings maturity, sensitivity to real interest rates, leverage and exposure to changes in risk premia. Such analysis would help determine whether relative repricing is primarily associated with financial duration or instead reflects sector-specific cash-flow news and other common shocks. More broadly, incorporating financial duration into the study of European monetary policy would improve the understanding of how Eurosystem interventions affect not only the level of asset prices, but also their internal and intertemporal composition.

9.4 Future Areas of Research

The first future line of research involves improving the Growth versus Value proxy used in this study. The use of the MSCI Europe Growth and MSCI Europe Value indices is justifiable because they capture

the cross-sectional dimension of financial duration, but they do not strictly represent the eurozone. A natural extension would be to construct or access MSCI EMU Growth and MSCI EMU Value series with sufficient coverage, homogeneous frequency and consistent methodology. This would allow testing the central hypothesis within a geographic sample more strictly aligned with the ECB's monetary policy.

The second line involves refining the measurement of financial duration. Growth and Value are useful but imperfect proxies. A more precise alternative would be to estimate implied duration at the firm or sector level using information on expected earnings, dividends, cash flows, valuation multiples, sensitivity to real interest rates and the term structure of expectations. Using microdata, one could estimate whether firms with higher implied duration respond more strongly to liquidity innovations, controlling for size, sector, profitability, leverage, intangibles and international exposure.

A third future line of research would be to expand the identification of monetary policy shocks. The use of residual liquidity innovations improves interpretation compared to simple correlations, but does not amount to strong exogenous identification. Future work could incorporate high-frequency identification around ECB announcements, monetary surprises derived from OIS, unexpected changes in interest rate expectations, or narrative strategies based on specific monetary policy decisions. This would allow for a more precise distinction between anticipated liquidity, unexpected liquidity, signaling regarding future rates and risk premium compression. In parallel, the uncertainty associated with the estimated first-stage reaction function could be propagated into the second-stage inference through bootstrap, joint estimation or resampling procedures, rather than treating the residual liquidity innovation as if it were observed without estimation error (Pagan, 1984).

The fourth line of research consists of better separating the transmission channels. Liquidity can affect asset prices through the risk-free rate, risk premiums, liquidity premiums, earnings expectations, portfolio rebalancing, and reduced financial fragmentation. This paper groups these mechanisms under a general framework of discount rates and risk premiums. A more ambitious extension could empirically decompose these channels using models of term structure of interest rates, corporate spreads, measures of sovereign risk, market liquidity indicators and earnings expectations. A further methodological extension would replace exclusively pointwise inference with joint inference across horizons, using simultaneous confidence bands or multi-horizon tests to characterise the response profile as a whole rather than isolated horizon-specific coefficients (Montiel Olea & Plagborg-Møller, 2019).

The fifth future line of research concerns regime dependence. The results suggest some heterogeneity, but do not allow us to conclude that there is robust nonlinearity. Future work could employ more sophisticated methods to identify financial regimes, such as Markov-switching models, threshold regressions, smooth transition models, or classifications based on multiple stress indicators. It would also be useful to distinguish between different types of stress: banking stress, sovereign stress, stock market stress, liquidity stress and macroeconomic stress. Not all of these episodes necessarily generate the same monetary transmission.

The sixth line of research involves studying the sectoral dimension. Growth and Value do not differ solely by financial duration but also by sectoral composition. Part of the differential response may reflect exposure to technology, healthcare, consumer discretionary, banking, energy, or industrial sectors. A sectoral extension would allow us to analyze whether relative outperformance is concentrated in sectors with higher implied duration or whether it responds to composition factors not directly linked to the discount channel. This distinction would be essential to strengthen the interpretation of financial

duration.

The seventh future line of research would be to connect the financial evidence with real variables of capital allocation. This paper analyzes relative prices, not real investment. However, if liquidity persistently favors assets with higher financial duration, a subsequent question is whether this affects investment allocation, corporate financing, equity issuance, R&D spending, mergers and acquisitions, or the sectoral structure of growth. This extension would allow us to move from financial valuation to the intertemporal coordination of the real economy.

In summary, this paper offers an initial empirical analysis of the relationship between European liquidity architecture and the relative valuation of assets with different financial durations. Its main contribution is to demonstrate that Eurosystem liquidity can be analyzed not only as an aggregate stabilization variable but also as a mechanism potentially linked to the intertemporal structure of prices. The evidence obtained does not settle the matter, but it opens a relevant avenue of research into how European monetary policy influences the valuation of time, risk, and future cash flows in financial markets.

References

Adrian, T., & Shin, H. S. (2010). Liquidity and leverage. *Journal of Financial Intermediation*, *19*(3), 418–437. https://doi.org/10.1016/j.jfi.2008.12.002

Altavilla, C., Carboni, G., & Motto, R. (2015). *Asset purchase programmes and financial markets: Lessons from the euro area* (Working Paper Series No. 1864). European Central Bank. https://www.ecb.europa.eu/pub/pdf/scpwps/ecbwp1864.en.pdf

Bauer, M. D., & Rudebusch, G. D. (2014). The signaling channel for federal reserve bond purchases. *International Journal of Central Banking*, *10*(3), 233–289.

Borio, C. (2014). The financial cycle and macroeconomics: What have we learnt? *Journal of Banking & Finance*, *45*, 182–198. https://doi.org/10.1016/j.jbankfin.2013.07.031

Borio, C., & Zhu, H. (2012). Capital regulation, risk-taking and monetary policy: A missing link in the transmission mechanism? *Journal of Financial Stability*, *8*(4), 236–251. https://doi.org/10.1016/j.jfs.2011.12.003

Campbell, J. Y., & Shiller, R. J. (1988). The dividend-price ratio and expectations of future dividends and discount factors. *The Review of Financial Studies*, *1*(3), 195–228. https://doi.org/10.1093/rfs/1.3.195

Campbell, J. Y., & Vuolteenaho, T. (2004). Bad beta, good beta. *American Economic Review*, *94*(5), 1249–1275. https://doi.org/10.1257/0002828043052240

Cieslak, A., & Vissing-Jorgensen, A. (2021). The economics of the fed put. *The Review of Financial Studies*, *34*(9), 4045–4089. https://doi.org/10.1093/rfs/hhaa116

Cochrane, J. H. (2005). *Asset pricing* (Revised edition). Princeton University Press.

Curvo. (2026a). Historical performance of the msci europe growth index in eur [Monthly data manually downloaded on 2026-05-29].

Curvo. (2026b). Historical performance of the msci europe value index in eur [Monthly data manually downloaded on 2026-05-29].

Dechow, P. M., Sloan, R. G., & Soliman, M. T. (2004). Implied equity duration: A new measure of equity risk. *Review of Accounting Studies*, *9*(2), 197–228. https://doi.org/10.1023/B:RAST.0000028186.44328.3f

Eisenschmidt, J., Kedan, D., & Tietz, R. (2018). Measuring fragmentation in the euro area unsecured overnight interbank money market: A monetary policy transmission approach. *ECB Economic*

Bulletin, (5). https://www.ecb.europa.eu/press/economic-bulletin/articles/2018/html/ecb.ebart201805_02.en.html

European Central Bank. (2022). The transmission protection instrument [Accessed 2026-05-30]. https://www.ecb.europa.eu/press/pr/date/2022/html/ecb.pr220721~973e6e7273.en.html

European Central Bank. (2023). What is excess liquidity? [Accessed 2026-05-30]. https://www.ecb.europa.eu/ecb-and-you/explainers/tell-me-more/html/excess_liquidity.en.html

European Central Bank. (2024a). Changes to the operational framework for implementing monetary policy [Accessed 2026-05-30]. https://www.ecb.europa.eu/press/pr/date/2024/html/ecb.pr240313~807e240020.en.html

European Central Bank. (2024b). The eurosystem's instruments [Accessed 2026-05-30]. https://www.ecb.europa.eu/mopo/implement/html/index.en.html

European Central Bank. (2024c). Open market operations [Accessed 2026-05-30]. https://www.ecb.europa.eu/mopo/implement/omo/html/index.en.html

European Central Bank. (2026a). Asset purchase programmes [Accessed 2026-05-30]. https://www.ecb.europa.eu/mopo/implement/app/html/index.en.html

European Central Bank. (2026b). Ecb data portal [Accessed 2026-05-30]. https://data.ecb.europa.eu/

European Central Bank. (2026c). Pandemic emergency purchase programme [Accessed 2026-05-30]. https://www.ecb.europa.eu/mopo/implement/pepp/html/index.en.html

Gagnon, J., Raskin, M., Remache, J., & Sack, B. (2011). Large-scale asset purchases by the federal reserve: Did they work? *Federal Reserve Bank of New York Economic Policy Review*, *17*(1), 41–59.

Giglio, S., Kelly, B., & Kozak, S. (2024). Equity term structures without dividend strips data. *The Journal of Finance*, *79*(6), 4143–4196. https://doi.org/10.1111/jofi.13394

Gormsen, N. J., & Lazarus, E. (2026). *Interest rates and equity valuations* (Working Paper No. 34814). National Bureau of Economic Research. https://doi.org/10.3386/w34814

Gürkaynak, R. S., Sack, B., & Swanson, E. T. (2005). Do actions speak louder than words? the response of asset prices to monetary policy actions and statements. *International Journal of Central Banking*, *1*(1), 55–93.

Hansen, L. P., & Hodrick, R. J. (1980). Forward exchange rates as optimal predictors of future spot rates: An econometric analysis. *Journal of Political Economy*, *88*(5), 829–853. https://doi.org/10.1086/260910

Holló, D., Kremer, M., & Lo Duca, M. (2012). *Ciss: A composite indicator of systemic stress in the financial system* (Working Paper Series No. 1426). European Central Bank.

Jiménez, G., Ongena, S., Peydró, J.-L., & Saurina, J. (2014). Hazardous times for monetary policy: What do twenty-three million bank loans say about the effects of monetary policy on credit risk-taking? *Econometrica*, *82*(2), 463–505. https://doi.org/10.3982/ECTA10104

Jordà, Ò. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review*, *95*(1), 161–182. https://doi.org/10.1257/0002828053828518

Krishnamurthy, A., & Vissing-Jorgensen, A. (2011). The effects of quantitative easing on interest rates: Channels and implications for policy. *Brookings Papers on Economic Activity*, *2011*(2), 215–287.

Lettau, M., & Wachter, J. A. (2007). Why is long-horizon equity less risky? a duration-based explanation of the value premium. *The Journal of Finance*, *62*(1), 55–92. https://doi.org/10.1111/j.1540-6261.2007.01201.x

Montiel Olea, J. L., & Plagborg-Møller, M. (2019). Simultaneous confidence bands: Theory, implementation, and an application to SVARs. *Journal of Applied Econometrics*, *34*(1), 1–17. https://doi.org/10.1002/jae.2656

MSCI. (2021). Msci global investable market value and growth index methodology [Accessed 2026-05-30]. https://www.msci.com/eqb/methodology/meth_docs/MSCI_GIMIVGMethod_Feb2021.pdf

MSCI. (2026a). Msci europe growth index [Accessed 2026-05-30]. https://www.msci.com/indexes/index/105843

MSCI. (2026b). Msci europe value index [Accessed 2026-05-30]. https://www.msci.com/indexes/index/105844

Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, *55*(3), 703–708. https://doi.org/10.2307/1913610

Pagan, A. (1984). Econometric issues in the analysis of regressions with generated regressors. *International Economic Review*, *25*(1), 221–247. https://doi.org/10.2307/2648877

Swanson, E. T. (2021). Measuring the effects of federal reserve forward guidance and asset purchases on financial markets. *Journal of Monetary Economics*, *118*, 32–53. https://doi.org/10.1016/j.jmoneco.2020.09.003

Vayanos, D., & Vila, J.-L. (2021). A preferred-habitat model of the term structure of interest rates. *Econometrica*, *89*(1), 77–112. https://doi.org/10.3982/ECTA17440

A. Appendices

A.1 Variable Dictionary

All variables used or described in the empirical analysis are listed below. Definitions, transformations and limitations are developed in Chapters 5 and 6, this table provides a concise traceability reference. MSCI EMU Growth/Value and European credit spreads are discussed as future extensions in section 9.4 and are not listed here.

Table 3
Variable dictionary

Variable	Definition	Source	Original frequency	Transformation	Effective sample	Empirical role
Eurosystem liquidity						
L_t	Total Eurosystem assets, EUR millions	ECB ILM, `W.U2.C.T000000.Z5.Z01`	Weekly	$\ell_t = \log L_t$; $g_t^L = 100\,\Delta\ell_t$	2000M1–2026M2	Baseline balance-sheet measure; dependent variable in the reaction function; source series used to construct $\Delta\ell_t$ and $\tilde{\varepsilon}_t^{\ell}$.
EL_t	Excess liquidity, defined as reserves minus minimum reserve requirements	ECB ILM, `D.U2.C.EXLIQ.U2.EUR`	Daily	$\log(EL_t)$; logarithmic difference	2024M9–2026M2, 18 observations	Available but excluded from the baseline specifications because of insufficient time coverage.
APP/PEPP	Programme-specific asset holdings	ECB Data Portal	Not established	Not used	Not used	Institutional context only; series identifiers and aggregation definitions remain unconfirmed[a].
Financial conditions						
CISS	Composite Indicator of Systemic Stress (Holló et al., 2012)	ECB CISS, `D.U2.Z0Z.4F.EC.SS_CIN`	Daily	Level; $\Delta_h CISS_{t+h} = CISS_{t+h} - CISS_{t-1}$	2000M1–2026M2	Reaction-function control; stress-regime criterion; dependent variable in the H2 specifications.
VSTOXX	Implied volatility index associated with the EURO STOXX 50	Yahoo Finance, `V2TX.DE`, unconfirmed	Daily	Not used	Not available	Intended equity-volatility control; absent from all estimated specifications[b].
Monetary policy and term structure						
i_t^{DFR}	ECB deposit facility rate, percentage points	ECB FM, `D.U2.EUR.4F.KR.DFR.LE`	Daily	Δi_t^{DFR}	2000M1–2026M2	Monetary-policy control included in the Eurosystem reaction function.
€STR	Euro short-term rate, percentage points	ECB EST, `B.EU000A2X2A25.WT`	Business daily	Δ €STR	2019M10–2026M2	Excluded from the baseline specifications because its available history is too short for the full sample.
i_t^{2Y}, i_t^{10Y}	Euro-area government yield-curve spot rates at two- and ten-year maturities	ECB YC, `B.U2.EUR.4F.G_N_C.SV_` and `SR_10Y`	Business daily	Δi_t^{2Y}, Δi_t^{10Y}; slope$_t = i_t^{10Y} - i_t^{2Y}$	2004M9–2026M2, 258 observations	Construction of the yield-curve slope and additional control in the Eurosystem reaction function[c].
Equity market						
EURO STOXX 50	Euro-area large-cap equity price index, denominated in euros	ECB RTD, `M.S0.N.C_DJE50.X`	Monthly	$r_t^M = \log P_t^M - \log P_{t-1}^M$	2000M1–2026M2	Aggregate equity-market control; dependent variable in the H2 specifications; base series for monthly data alignment.
Growth versus Value						

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Variable	Definition	Source	Original frequency	Transformation	Effective sample	Empirical role
r_t^G, r_t^V	Monthly logarithmic returns on MSCI Europe Growth and MSCI Europe Value in euros	Curvo (Curvo, 2026a, 2026b)	Monthly	$r_t = \log P_t - \log P_{t-1}$	Empirical sample: 2000M1–2026M2	Components used to construct the H4 dependent variable[d].
GV_t	$r_t^G - r_t^V$; positive values indicate Growth outperformance	Derived	Monthly	$GV_{t,t+h} = \sum_{j=0}^{h} \left(r_{t+j}^G - r_{t+j}^V \right)$	2000M1–2026M2	Central dependent variable in the H4 local-projection specifications.
Derived variables						
$\tilde{\varepsilon}_t^{\ell}$	Standardised residual from the Eurosystem liquidity reaction function	Derived	Monthly	$\tilde{\varepsilon}_t^{\ell} = \frac{\hat{\varepsilon}_t^{\ell}}{\hat{\sigma}_{\varepsilon^{\ell}}}$	$N = 254$	Baseline liquidity-innovation variable used in the local projections; alternative liquidity measures are considered in the robustness specifications[e].
D_t^{Stress}	$\mathbf{1}\{CISS_{t-1} > Q_{0.75}\}$; threshold 0.2335; 78 stress observations and 235 normal-regime observations	Derived	Monthly	Binary indicator	2000M2–2026M2, 313 observations	Interaction variable used in the regime-dependent H3 specifications.

Note. [a] APP and PEPP series identifiers and aggregation definitions were not confirmed; the programmes are therefore retained as institutional context only. [b] The attempted Yahoo Finance download using `V2TX.DE` produced no usable observations. [c] The 258 observations reflect the joint availability of the two- and ten-year yield series after monthly aggregation. [d] The raw Curvo series run from December 1974 to April 2026; the empirical sample is restricted to January 2000 through February 2026. [e] The reaction function is estimated on 254 observations after applying three lags.

Source: Own elaboration.

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A.2 Data Sources and Traceability

All ECB series were downloaded via the ECB Data Portal (https://data.ecb.europa.eu/) (European Central Bank, 2026b) using the flow and key identifiers confirmed in `config.py`. Monthly aggregation follows the end-of-period convention (Section 5.2). MSCI Europe Growth and Value series were obtained from Curvo as third-party historical performance series representing those indices in EUR, downloaded manually on 2026-05-29 (Curvo, 2026a, 2026b).

Table 4

Data sources and traceability

Series	Provider	Flow and key	Coverage	Status	Relevant limitation
Eurosystem total assets	ECB Data Portal	ILM: `W.U2.C.T000000.Z5.Z01`	2000M1–2026M2	Operative	Broad balance-sheet measure that does not isolate individual refinancing or asset-purchase programmes.
Excess liquidity	ECB Data Portal	ILM: `D.U2.C.EXLIQ.U2.EUR`	2024M9–2026M2, 18 observations	Not in baseline	Coverage is insufficient for estimation over the full sample.
APP/PEPP holdings	ECB Data Portal	Identifiers not confirmed[a]	Not incorporated	Not incorporated	Definitions and aggregation procedures were not sufficiently verified.
CISS	ECB Data Portal	CISS: `D.U2.Z0Z.4F.EC.SS_CIN.IDX`	2000M1–2026M2	Operative	System-wide indicator that does not isolate equity-market volatility.
VSTOXX	Yahoo Finance	Attempted ticker: `V2TX.DE`[b]	Unconfirmed	Not available	The attempted download produced no usable observations; the variable is absent from all specifications.
Deposit facility rate	ECB Data Portal	FM: `D.U2.EUR.4F.KR.DFR.LEV`	2000M1–2026M2	Operative	The level takes negative values during part of the sample; simple first differences are therefore used.
STR	ECB Data Portal	EST: `B.EU000A2X2A25.WT`	2019M10–2026M2	Not in baseline	The available history is too short for the full-sample specification.
Two-year yield	ECB Data Portal	YC: `B.U2.EUR.4F.G_N_C.SV_C_YM.SR_2Y`	2004M9–2026M2	Operative	Euro-area government yield-curve spot rate; joint availability with the ten-year series gives 258 observations.
Ten-year yield	ECB Data Portal	YC: `B.U2.EUR.4F.G_N_C.SV_C_YM.SR_10Y`	2004M9–2026M2	Operative	Used together with the two-year rate to construct the yield-curve slope.
EURO STOXX 50	ECB Data Portal	RTD: `M.S0.N.C_DJE50.X`	2000M1–2026M2	Operative	Native monthly price index; returns exclude any separate dividend adjustment not embedded in the source series.
MSCI Europe Growth, EUR	Curvo (Curvo, 2026a)	Third-party historical performance series representing MSCI Europe Growth; manually downloaded on 2026-05-29	Raw: 1974M12–2026M4; empirical: 2000M1–2026M2	Operative	Not downloaded directly from MSCI; includes developed European markets outside the euro area.
MSCI Europe Value, EUR	Curvo (Curvo, 2026b)	Third-party historical performance series representing MSCI Europe Value; manually downloaded on 2026-05-29	Raw: 1974M12–2026M4; empirical: 2000M1–2026M2	Operative	Same geographical limitation as the Growth series; it is not a pure euro-area portfolio.

Note. [a] APP and PEPP are retained only as institutional context because their series identifiers and aggregation definitions were not confirmed. [b] The attempted VSTOXX download using the Yahoo Finance ticker `V2TX.DE` produced no usable observations; its source and coverage therefore remain unconfirmed.

Source: Own elaboration.

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A.3 Mathematical Transformations

For any positive index P_t, the monthly logarithmic return is defined as:

$$r_t = \log(P_t) - \log(P_{t-1})$$

When expressed in approximate percentage points:

$$R_t = 100[\log(P_t) - \log(P_{t-1})]$$

The cumulative return between t and $t + h$ is calculated as:

$$R_{t,t+h} = \sum_{j=0}^{h} r_{t+j}$$

For positive liquidity variables L_t, the following is defined:

$$\ell_t = \log(L_t)$$

and its monthly growth, as follows:

$$\Delta\ell_t = \ell_t - \ell_{t-1}$$

If expressed in approximate percentage points:

$$g_t^L = 100(\ell_t - \ell_{t-1})$$

For interest rates and spreads expressed as percentages, simple calculations are used:

$$\Delta i_t = i_t - i_{t-1}$$

The slope of the curve is calculated as:

$$slope_t = i_t^{10Y} - i_t^{2Y}$$

and its monthly change as:

$$\Delta slope_t = slope_t - slope_{t-1}$$

The central convention of the cross-party bloc is:

$$GV_t = r_t^{Growth} - r_t^{Value}$$

Under this convention, $GV_t > 0$ indicates that Growth outperforms Value. If the inverse convention is used,

$$VG_t = r_t^{Value} - r_t^{Growth}$$

Therefore:

$$VG_t = -GV_t$$

Thus, the expected sign of the empirical hypothesis is reversed. To avoid ambiguity, the main text, figures, and tables always use “Growth minus Value.”

A.4 Formal Derivation of Financial Duration and Growth vs. Value

This appendix formally elaborates on the relationship between intertemporal valuation, financial duration, and sensitivity to the discount rate. The aim is to provide a mathematical justification for the approach used in the main body of the paper without overcomplicating the theoretical framework.

We start with the simplified expression for the value:

$$P_t = \mathbb{E}_t \left[\sum_{j=1}^{\infty} \frac{CF_{t+j}}{(1+R_t)^j} \right]$$

where P_t represents the asset price, CF_{t+j} the expected cash flow at $t+j$, and R_t the aggregate discount rate. Differentiating with respect to R_t, while holding cash flow expectations constant, we obtain:

$$\frac{\partial P_t}{\partial R_t} = -\mathbb{E}_t \left[\sum_{j=1}^{\infty} \frac{jCF_{t+j}}{(1+R_t)^{j+1}} \right]$$

Since the expected cash flows are positive in a standard equity valuation model, the derivative is negative. Therefore, a decrease in the discount rate increases the asset’s price, while an increase in the discount rate decreases it.

We define the relative weight of the expected cash flow at horizon j as:

$$w_{t,j} = \frac{\mathbb{E}_t \left[\frac{CF_{t+j}}{(1+R_t)^j} \right]}{P_t}$$

If P_t is the sum of the expected present values of all cash flows, then:

$$\sum_{j=1}^{\infty} w_{t,j} = 1$$

The asset’s implied Macaulay duration is defined as:

$$D_t^M = \sum_{j=1}^{\infty} j w_{t,j}$$

This expression measures the weighted average time horizon of the cash flows that support the asset’s value. The greater the weight of future cash flows, the larger D_t^M will be.

Based on the previous derivative, the proportional sensitivity of the price to the discount rate is:

$$\frac{1}{P_t} \frac{\partial P_t}{\partial R_t} = -\frac{\mathbb{E}_t \left[\sum_{j=1}^{\infty} \frac{jCF_{t+j}}{(1+R_t)^{j+1}} \right]}{P_t}$$

Rearranging the term $(1+R_t)^{-(j+1)}$ as $(1+R_t)^{-1}(1+R_t)^{-j}$, we obtain:

$$\frac{1}{P_t}\frac{\partial P_t}{\partial R_t} = -\frac{1}{1+R_t}\sum_{j=1}^{\infty} j \frac{\mathbb{E}_t\left[\frac{CF_{t+j}}{(1+R_t)^j}\right]}{P_t}$$

Using the definition of $w_{t,j}$, we get:

$$\frac{1}{P_t}\frac{\partial P_t}{\partial R_t} = -\frac{1}{1+R_t}\sum_{j=1}^{\infty} j w_{t,j}$$

thus:

$$\frac{\partial \ln P_t}{\partial R_t} = -\frac{D_t^M}{1+R_t}$$

We define the modified duration as:

$$D_t = \frac{D_t^M}{1+R_t}.$$

then:

$$\frac{\partial \ln P_t}{\partial R_t} = -D_t$$

For small changes in the discount rate, the first-order approximation can be used:

$$\Delta \ln P_t \approx -D_t \Delta R_t$$

This expression implies that the percentage change in price resulting from a change in the discount rate is proportional to the financial life of the asset. If $\Delta R_t < 0$, then $\Delta \ln P_t > 0$. Furthermore, the larger D_t is, the greater the percentage increase in price.

Let us now consider two portfolios, *G* and *V*, where *G* represents a *Growth* portfolio and *V* a *Value* portfolio. If the *Growth* portfolio has a longer financial duration than the *Value* portfolio, then:

$$D_t^G > D_t^V$$

Given a common or highly correlated variation in the discount rate, the approximate price changes are:

$$\Delta \ln P_t^G \approx -D_t^G \Delta R_t$$

$$\Delta \ln P_t^V \approx -D_t^V \Delta R_t$$

Subtracting both expressions:

$$\Delta \ln P_t^G - \Delta \ln P_t^V \approx -\left(D_t^G - D_t^V\right)\Delta R_t$$

if $D_t^G - D_t^V > 0$ and $\Delta R_t < 0$, then:

$$\Delta \ln P_t^G - \Delta \ln P_t^V > 0$$

This disparity forms the basis of the *Growth versus Value* hypothesis. If an expansion of liquidity compresses the aggregate discount rate, and if *Growth* has a longer financial duration than *Value*, then *Growth* should experience a relative appreciation relative to *Value*.

The empirical variable used to capture this relationship is:

$$GV_t = r_t^{Growth} - r_t^{Value}$$

Under this convention, $GV_t > 0$ indicates that *Growth* outperforms *Value* in period t. For cumulative horizons:

$$GV_{t,t+h} = \sum_{j=0}^{h} \left(r_{t+j}^{Growth} - r_{t+j}^{Value} \right)$$

The empirical hypothesis associated with the local projections is:

$$GV_{t,t+h} = \alpha_h + \beta_h \varepsilon_t^{\ell} + \Gamma_h' X_{t-1} + u_{t+h}$$

with:

$$\beta_h > 0$$

in scenarios where expansive liquidity innovation is transmitted to relative valuations through a compression of discount rates or risk premiums.

A.5 General Structure of the Dataset

Table 5
General structure of the dataset

Block	Key variables	Empirical function
Eurosystem liquidity	Eurosystem total assets, excess liquidity, APP holdings, and PEPP holdings, where available	Align balance-sheet expansion with central-bank liquidity provision.
Financial conditions	CISS, VSTOXX, and credit spreads, where available	Capture systemic stress, market volatility, and initial financial conditions.
Monetary policy and interest rates	Deposit facility rate, STR, two-year yield, ten-year yield, and ten-year–two-year spread	Assess the monetary policy stance and the term structure of interest rates.
Aggregate equity market	EURO STOXX 50 and its monthly return	Measure the response of the aggregate euro-area equity market.
Growth versus Value	MSCI Europe Growth, MSCI Europe Value, and Growth minus Value	Approximate relative financial duration and sensitivity to changes in discount rates.

Source: Own elaboration.

A.6 Supplementary Econometric Results

A.6.1 Eurosystem Reaction Function and Predictable Component

Table 6

Eurosystem liquidity reaction function

Variable	Coefficient	HAC standard error	HAC p-value	Signifi-cance
Constant	0.0057	0.0023	0.014	**
Growth in Eurosystem assets, $t-3$	0.2582	0.061	< 0.001	***
EURO STOXX 50 return, $t-3$	-0.1323	0.0629	0.036	**
Change in the deposit facility rate, $t-3$	-0.0315	0.0176	0.075	*
Change in the 10-year–2-year yield spread, t − 3, $t-3$	-0.0251	0.0155	0.107	–

Statistics of the preferred model, $p = 3$	Value
N	254
adjusted R^2	0.1494
AIC	-1064.997
BIC	-1008.400
Durbin–Watson	1.954
Ljung–Box p-value lag 12	0.237
Jarque–Bera p-value	< 0.001
Breusch–Pagan p-value	< 0.001

Note. HAC/Newey–West standard errors are reported. Asterisks indicate $p < 0.01$ (***), $p < 0.05$ (**), and $p < 0.10$ (*). The model residual is interpreted as a reduced-form proxy for liquidity innovation rather than as a structural monetary policy shock.

Source: Own elaboration.

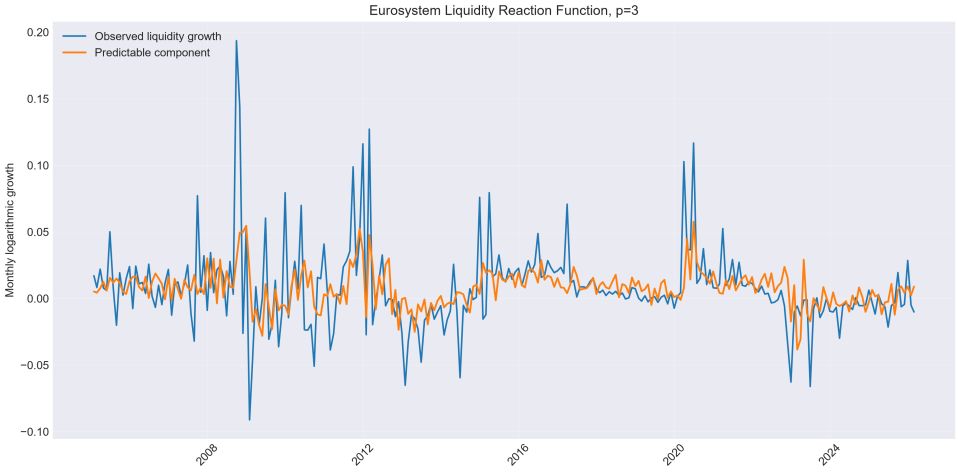


Figure 5

Observed growth and the predictable component of Eurosystem liquidity

Source: Author's own calculations and estimates based on ECB Data Portal data (European Central Bank, 2026b).

A.6.2 Regime-Dependent Estimates

Table 7

Estimated responses by financial stress regime

Horizon	Normal response	High-stress response	Difference
Panel A. EURO STOXX 50 cumulative response, percentage points			
$h = 0$	-0.842	-0.717	0.126
$h = 6$	-0.252	-3.015	-2.763
$h = 12$	-2.432	-0.730	1.702
$h = 24$	-1.202	-1.482	-0.280
Panel B. CISS response in levels			
$h = 0$	0.0279	0.0165	-0.0114
$h = 6$	-0.0014	0.0413	0.0427
$h = 12$	0.0301	-0.0036	-0.0337
$h = 24$	-0.0088	0.0055	0.0143

Note. High stress is defined as $CISS_{t-1}$ above its 75th percentile. The corresponding threshold is 0.2335, and high-stress observations represent approximately 24.9% of the sample. Panel A reports cumulative responses in percentage points, whereas Panel B reports responses in CISS units. The difference is calculated as the high-stress response minus the normal-regime response.

Source: Own elaboration.

A.6.3 Supplementary Tables: Growth vs. Value Block

Table 8

Baseline local projection of Growth minus Value by horizon

h	β_h	Standard error	*t*	*p*-value	90% CI	95% CI	*N*
0	0.0014	0.0019	0.71	0.479	[-0.0018, 0.0045]	[-0.0024, 0.0051]	254
1	0.0038	0.0027	1.41	0.160	[-0.0006, 0.0083]	[-0.0015, 0.0092]	253
2	0.0035	0.0031	1.11	0.266	[-0.0017, 0.0087]	[-0.0027, 0.0097]	252
3	0.0058^{**}	0.0028	2.05	0.041	[0.0012, 0.0105]	[0.0003, 0.0114]	251
4	0.0087^{***}	0.0033	2.60	0.010	[0.0032, 0.0141]	[0.0021, 0.0152]	250
5	0.0092^{***}	0.0034	2.75	0.006	[0.0037, 0.0148]	[0.0026, 0.0158]	249
6	0.0082^{**}	0.0032	2.55	0.011	[0.0029, 0.0135]	[0.0019, 0.0145]	248
7	0.0070^{**}	0.0036	1.98	0.049	[0.0012, 0.0129]	[0.0001, 0.0140]	247
8	0.0060^{**}	0.0028	2.15	0.032	[0.0014, 0.0107]	[0.0005, 0.0115]	246
9	0.0070^{**}	0.0032	2.15	0.032	[0.0016, 0.0123]	[0.0006, 0.0133]	245
10	0.0047	0.0038	1.24	0.218	[-0.0016, 0.0110]	[-0.0028, 0.0122]	244
11	0.0014	0.0046	0.30	0.768	[-0.0062, 0.0089]	[-0.0076, 0.0103]	243
12	0.0023	0.0048	0.48	0.633	[-0.0056, 0.0102]	[-0.0071, 0.0117]	242

Note. The baseline specification uses a standardized residual liquidity innovation obtained from the Eurosystem reaction function and HAC/Newey–West standard errors. Coefficients are expressed in cumulative log-return units; multiplying them by 100 yields approximate percentage-point responses. The dependent variable is Growth minus Value, so a positive β_h indicates relative outperformance of Growth over Value. MSCI Europe Growth and MSCI Europe Value are used as proxies for the Growth-versus-Value dimension in developed European equity markets and should not be interpreted as MSCI EMU Growth and MSCI EMU Value portfolios. Asterisks indicate $p < 0.01$ (***), $p < 0.05$ (**), and $p < 0.10$ (*).

Source: Own elaboration.

Table 9

Specification summary for Growth minus Value

Specification	Peak β	Peak h	Average $h = 0$–3	Average $h = 4$–8	Average $h = 9$–12	Significant at 10%	Significant at 5%
Baseline liquidity innovation	0.0092^{***}	5	0.0036	0.0078	0.0038	7	7
Richer controls	0.0093^{***}	5	0.0036	0.0079	0.0043	7	7
Baseline, two lags	0.0102^{***}	5	0.0038	0.0083	0.0034	7	4
Baseline, three lags	0.0102^{***}	5	0.0037	0.0083	0.0035	7	5
Post-2009 sample	0.0110^{**}	7	0.0062	0.0092	0.0065	9	5
Post-2012 sample	0.0142^{***}	3	0.0101	0.0107	0.0054	5	4
Excluding COVID window	0.0081^{**}	5	0.0020	0.0067	0.0050	7	5
Balance-sheet growth	0.0101^{***}	4	0.0043	0.0084	0.0043	8	5
Reaction residual	0.0092^{***}	5	0.0036	0.0078	0.0038	7	7

Note. Coefficients are expressed in cumulative log-return units. Asterisks indicate $p < 0.01$ (***), $p < 0.05$ (**), and $p < 0.10$ (*), evaluated at the peak horizon. The dependent variable is Growth minus Value, so positive coefficients indicate relative outperformance of Growth over Value. The baseline liquidity innovation is the standardized residual liquidity innovation obtained from the Eurosystem reaction function. Alternative liquidity specifications use standardized transformations. MSCI Europe Growth and MSCI Europe Value are used as proxies for the Growth-versus-Value dimension in developed European equity markets and should not be interpreted as pure MSCI EMU portfolios.

Source: Own elaboration.

B. Declaración de uso de inteligencia artificial

This declaration constitutes the full disclosure of AI use in this work, as the Methodology does not contain a separate AI-use description. Generative AI tools were used exclusively as support for drafting, code debugging, structuring and bibliographic search. They were not used to generate empirical results, data, statistical estimates or conclusions. The research design, the econometric specification, the interpretation of results (including the distinction between compatible evidence, conditional association and structural causality) and the final responsibility for the work remain entirely with the author. All AI-assisted content was reviewed, verified and, where necessary, corrected by the author. No AI tool was used to fabricate references, data or results.

Tools, versions and purpose

Table 10
Use of digital tools

Tool	Purpose
ChatGPT	Conceptual and bibliographic support.
Claude	Code support for local projections, Newey–West HAC standard errors, and the Eurosystem reaction function; structuring of sections and terminological consistency; writing assistance for English academic style and grammar review of the author's own text.
DeepL	Targeted translation, for example, resumen and abstract.
Perplexity	Search for relevant bibliography.

Representative prompts

The following prompts illustrate how the tools were used. They were oriented towards critical review, verification and support, not towards generating the substantive content of the work.

1. *Critical review (examiner perspective).* "Act as a demanding TFG examiner in Economics and International Business. Assess the soundness of the research question, internal coherence, methodology, interpretation of results, quality of citations, structure and academic style. Point out concrete errors, penalisation risks and priority improvements."

2. *Internal consistency check.* "Review the document for internal inconsistencies across abstract, introduction, methodology, results, tables, figures, discussion and conclusions. Check whether reported figures match, whether hypotheses are evaluated consistently and whether textual claims are supported by the empirical results."

3. *Causal prudence review.* "Critically review the sections on intertemporal fragility, financial duration, Eurosystem liquidity and causality. Assess whether the claims are sufficiently supported by the methodology and the literature, and flag specific sentences that should be softened, replaced or reinforced with citations."

4. *Length and formatting optimisation.* "Identify minor adjustments in wording, formatting, redundancies, tables, figures or spacing that reduce page count without weakening the academic content. Prioritise safe, low-risk changes and avoid substantive cuts."

5. *Bibliographic support.* “Given the current state of the document, recommend additional relevant bibliography where concrete gaps are detected.”